

CPP**ELECTROSTATICS -SHEET: 1(Lecture – 1)****Level-I**

- How many electrons are required to be taken out of a body to give a net charge of $1 \mu\text{C}$ on it?
- Can a body have a stable charge of $2 \times 10^{-19} \text{ C}$ on it? Why?
- Find the charge on a copper sphere of mass 1 kg, if one electron is taken out from each of the copper atom. Atomic mass of Cu is 65.
- Which one of the following statement regarding electrostatics is wrong?
 - Charge is quantized
 - Charge is conserved
 - There is an electric field near an isolated charge at rest
 - A stationary charge produces both electric and magnetic fields
- Does charging involve transfer of mass?
- Which process is a sure test of nature of charge on bodies?
- When the distance between two charged particle is halved, the force between them becomes -
 - One fourth
 - One half
 - Double
 - Four times
- Two particles A and B having charges $8 \times 10^{-6} \text{ C}$ and $-2 \times 10^{-6} \text{ C}$ respectively are held fixed with a separation of 20 cm. Where should a third charged particle C be placed so that it does not experience a net electric force ?
- Two small spheres each having the charge $+Q$ are suspended by insulating threads of length L from a hook. This arrangement is taken in space where there is no gravitational effect, then the angle between the two suspensions and the tension in each will be

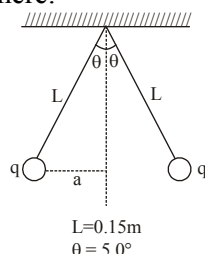
(A) $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{(2L)^2}$	(B) $90^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{L^2}$
(C) $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{2L^2}$	(D) $180^\circ, \frac{1}{4\pi\epsilon_0} \frac{Q^2}{L^2}$

Level-II

- Two small non-conducting spherical objects each of mass 5 kg carry the same amount of identical charges uniformly distributed over them. If the electrostatic force between the two objects is exactly balanced by gravitational force of attraction between them, find the magnitude of the charge (approximately).
- What would be the interaction force between two copper spheres, each of mass 1 g, separated by the distance 1 m, if the total electronic charge in them differed from the total charge of the nuclei by one per cent?
- A charge Q is divided into parts q and $Q - q$. Find the value of q for which the force between the two charges is maximum for a separation ' r ' between them.
- Two small positively charged spheres are suspended from a common point at the ceiling by the insulating light strings of equal length. The first sphere has mass m_1 and charge q_1 while the second one has mass m_2 and charge q_2 . If the first string makes an angle θ_1 with the vertical, find the angle θ_2 that the second string makes with the vertical.

5. Two conducting charged spheres of radius 'R' are kept at a distance 'd' ($d > 2R$). One has a charge $+q$ and the other $-q$. The electrostatic force between them will be
- (1) $\frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (2) $> \frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (3) $< \frac{1}{4\pi\epsilon_0} \frac{q^2}{d^2}$ (4) None of these

6. Two identical small charged spheres, each having a mass of 3.0×10^{-2} kg, hang in equilibrium as shown in figure. The length of each string is 0.15m, and the angle θ is 5.0° . Find the magnitude of the charge on each sphere.



7. If a negatively charged body is attracting a body of unknown nature of charged body. What are the possible signs of the unknown charged body. OR How charge and uncharged body attract?
8. Two spheres of the same metal (in all respects) are taken. One is given a positive charge of Q coulomb and other is given the same but negative charge. Which sphere will have a higher mass.
9. Which of the following is true:
- (i) Coulomb's law is applicable to point charges only. But it can be applied for distributed charges also.
 - (ii) Coulomb's law obeys Newton's third law
 - (iii) Electrostatic Force is conservative and central force.
 - (iv) Charge cannot exist without mass although mass can exist without charge
10. The electron and proton of a hydrogen atom are separated (on the average) by a distance of approximately 5.3×10^{-11} m. Find the magnitudes of the electric force and the gravitational force between the two particles.
11. What do you mean by "charge is invariant".

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ELECTROSTATICS -SHEET: 1(Lecture – 1)

ANSWER KEY

Level-I

1. 6.25×10^{12}
1. Charge $m \text{ e}^- = 1.6 \times 10^{-19} \text{ C}$
 $\Rightarrow 1 \text{ C} = \frac{1}{1.6 \times 10^{-19} \text{ e}^-}$
 $1 \mu\text{C} = \frac{1}{1.6 \times 10^{-19}} \times 10^{-6} \text{ e}^-$
 $= \frac{10}{1.6} = 6.25 \times 10^{12} \text{ e}^- \text{ s}$
2. No
2. No charge is quantized and can exist with stability if $q = ne$ if $n = \text{integer}$.
3. 1.48 MC
4. D

4. Theoretical
5. Yes
5. Yes because of transfer of charged particles.
6. Repulsion, not attraction
6. Theoretical
7. D

$$7. F \propto \frac{q_1 q_2}{r^2} \text{ if } r \text{ is solved}$$

$\Rightarrow F$ is four times.

$$8. x = 0.2 \text{ m}$$

8. As the net electric force on C should be equal to zero, the force due to A and B must be opposite in direction. Hence, the particle should be placed on the line AB. As A and B have charges of opposite signs, C cannot be between A and B. Also A has larger magnitude of charge than B. Hence, C should be placed closer to B than A. The situation is shown in figure. Suppose $BC = x$ and the charge on C is Q

$$\vec{F}_{CA} = \frac{1}{4\pi\epsilon_0} \frac{(8.0 \times 10^{-6}) Q}{(0.2 + x)^2} \hat{i}$$

$$\text{and } \vec{F}_{CB} = \frac{-1}{4\pi\epsilon_0} \frac{(2.0 \times 10^{-6}) Q}{x^2} \hat{i}$$

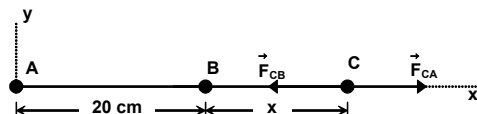
$$\vec{F}_C = \vec{F}_{CA} + \vec{F}_{CB}$$

$$= \frac{1}{4\pi\epsilon_0} \left[\frac{(8.0 \times 10^{-6}) Q}{(0.2 + x)^2} - \frac{(2.0 \times 10^{-6}) Q}{x^2} \right] \hat{i}$$

$$\text{But } |\vec{F}_C| = 0$$

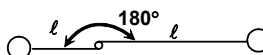
$$\text{Hence } \frac{1}{4\pi\epsilon_0} \left[\frac{(8.0 \times 10^{-6}) Q}{(0.2 + x)^2} - \frac{(2.0 \times 10^{-6}) Q}{x^2} \right] = 0$$

Which gives $x = 0.2 \text{ m}$



9. A
9. Is absence of gravity a due to repulsion = angle b/w threads = 180

$$T = \frac{\theta^2}{4\pi\epsilon_0 (2\ell)^2}$$



Level-II

$$1. \frac{q \cdot q}{4\pi\epsilon_0 r^2} = \frac{Gmm}{r^2}$$

$$q = m\sqrt{4\pi\epsilon_0 G} = 5\sqrt{\frac{6.67 \times 10^{-11}}{9 \times 10^9}} = 4.3 \times 10^{-4} \mu\text{C}$$

$$2. 2 \times 10^{15} \text{ N.}$$

$$F = \frac{kq_1 q_2}{r^2} = \frac{9 \times 10^9 \times 9^2}{r^2}$$

$$r = 1 \text{ N}$$

$$M = 1 \text{ gm} = \text{No. of in atom} = \frac{1}{65} \times 6.023 \times 10^{23}$$

$$\text{Charge of cn nucleus} = 29 e \text{ n } 1 \% \text{ of total charge} = \frac{1}{65} \times 6.023 \times 10 \times 29 \times 1.6 \times 10 \times \frac{1}{100} = q$$

$$q = 4.3 \times 4.3 \times 10^2 \text{ C}$$

$$F = 1.66 \times 10^{15} \text{ N } 2 \times 10^{15} \text{ N}$$

$$3. q = \frac{Q}{2}$$

$$3. \quad F = \frac{kq(\theta - q)}{r^2}$$

$$\text{for } F \text{ to be max } \frac{dF}{dq} = 0$$

$$\Rightarrow \frac{k}{r^2} [0 - 2 - q] = 0$$

$$\Rightarrow q = \frac{Q}{2}$$

$$4. \quad \frac{\sin \theta_1}{\sin \theta_2} = \frac{m_1}{m_2}$$

4. For m_2

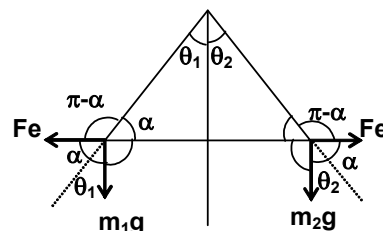
$$\frac{F_e}{\sin(\pi - \theta_2)} = \frac{m_2 g}{\sin(\pi - \alpha)} \quad (i)$$

For m_1

$$\frac{F_e}{\sin(\pi - \theta_1)} = \frac{m_1 g}{\sin(\pi - \alpha)} \quad (ii)$$

Divide (i) by (2)

$$\Rightarrow \frac{\sin \theta_1}{\sin \theta_2} = \frac{m_1}{m_2}$$



5. (2) Redistribution of charge will take place due to mutual attraction and hence effective distance will be less than d .

Note : In the example above, if both had the charge '+q', the answer would have been (3) because now mutual repulsion will result into increase in effective distance.

$$6. \quad (1) SF_x = T \sin q - F_e = 0$$

$$(2) SF_y = T \cos q - mg = 0$$

$$|q| = 4.4 \times 10^{-8} \text{ C}$$

There is no way we could find the sign of the charge from the information given. In fact, the sign of the charge is not important. The situation will be exactly the same whether both spheres are positively charged or negatively charged.

7. It can have any charge as a -ve charged body can definitely attract +ve, charged body, It can as well attract neutral or -ve charged body because of induction positive charge on a neutral body will be closer to the -ve charge. Although the charge on +ve body should be small so that effect of the distance overrule the slight effect of repulsion.

8. Negatively charged sphere will have a higher mass. This is due to increase in number of electron to make it negatively charged.

9. BCD

10. Charge and mass of the electron and proton

Particle	Charge (C)	Mass (kg)
Electron (e)	$-1.6021917 \times 10^{-19}$	9.1095×10^{-31}
Proton (P)	$+1.602191 \times 10^{-19}$	1.67261×10^{-27}

From Coulomb's law,

$$F_e = k_e = (8.99 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2) = 8.2 \times 10^{-8} \text{ N}$$

Using Newton's law of universal gravitation

$$F_g = G = (6.67 \times 10^{-11} \text{ N} \cdot \text{m}^2 / \text{kg}^2) \times = 3.6 \times 10^{-47} \text{ N}$$

$$\text{The ratio } F_e / F_g = 2 \times 10^{39}$$

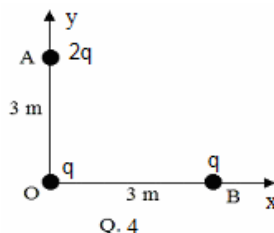
11. Charge is independent of frame of reference (unlike charge density and mass depend on frame of reference and increases with increase in speed)

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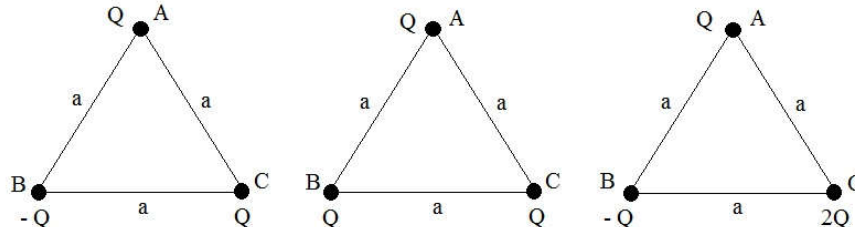
ELECTROSTATICS -SHEET: 2(Lecture – 2)

Level-I

- The permittivity of vacuum is $8.86 \times 10^{-12} \text{ C}^2/\text{N-m}^2$ and the dielectric constant of water is 81. Find permittivity of water in $\text{C}^2/\text{N-m}^2$
- Two point charged spheres separated at a distance d exert a force F on each other. If they are immersed in a liquid of dielectric constant 2, then what is the force that spheres exert on each other (if all conditions are same).
- $\sqrt{3} \times 10^{-19} \text{ C}$ and -10^{-6} C are placed at $(0, 0, 0)$ and $(1, 1, 1)$ respectively. Find the force on second in vector form.
- Find the unit vector in the direction of net electrostatic force on a charge q placed at the origin due to other two charges at A and B. $OA = OB = 3 \text{ m}$.



- Find the net electrostatic force on the charge at vertex A in following cases:



- The diagram shows four systems of charged particles. Each system contains two charged particles at point A and point B as shown in the figure. A third positive point charge q_0 is kept at point P in each system. The direction of electrostatic force experienced by the charge q_0 will be

direction of electric field lines experienced by the charge q_0 will be

System -1; $AP = BP = r$ and AB is a straight line

System -2; $AP = BP = r$ and AB is a straight line

System -3; $AP = BP = r$ and AB is a circular arc of radius r and centre at O

System -4; $AP = BP = r$ and AB is a circular arc of radius r and centre at O

System -1

System-2

System-3

System-4

(A)

$\uparrow$

$\rightarrow$

$\leftarrow$

$\downarrow$

(B)

$\downarrow$

$\leftarrow$

$\rightarrow$

$\uparrow$

(C)

$\uparrow$

$\rightarrow$

$\rightarrow$

$\downarrow$

(D)

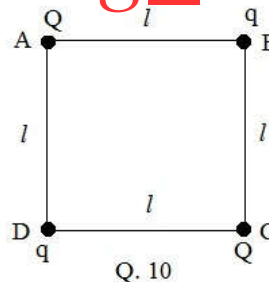
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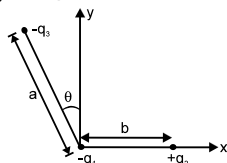
$\leftarrow$

$\uparrow$

7. Four point charges A, B, C, and D of charges Q , q , Q and q respectively are kept at the vertices of a square of side ' l '. Find $\frac{q}{Q}$ if net force acting on Q is zero.



8. Two similar charge of $+Q$, as shown in figure are placed at A and B. $-q$ charge is placed at point C midway between A and B. $-q$ charge will oscillate if
- (A) It is moved towards A (B) It is moved towards B
- (C) It is moved perpendicular to AB (D) Distance between A and B is reduced
9. Three charges $-q_1$, $+q_2$ and $-q_3$ are placed as shown in figure. What is the x-component of the force on $-q_1$.



LEVEL - II

- Four identical point charges each of magnitude q are placed at the corners of a square of side a . Find the net electrostatic force on any of the charge.
- Five equal charges ' q ' are placed at five vertices of a regular hexagon. What will be the resultant force on a charge ' Q ' placed at the centre of the hexagon given that the distance of a corner from centre is d .
- Two positive point charges of magnitude Q each are separated by a distance " $2a$ ". A test charge ' q_0 ' is located in a plane which is normal to the line joining these charges and midway between them. Find the locus of the points in this plane for which the force on the test charge has a maximum value.
- A thin wire ring of radius r has an electric charge q . What will be the increment of the force stretching the wire if a point charge q_0 is placed at the ring's centre?
- Two free point charges $+4Q$ and $+Q$ are placed at a distance r . A third charge q is so placed that all the three are in equilibrium.

(A) q is placed at a distance $\frac{1}{3}r$ from $4Q$ (B) q is placed at a distance $\frac{1}{3}r$ from Q

(C) $q = \frac{4Q}{9}$ (D) $q = -\frac{4Q}{9}$
- Two point charges repel each other with a force of 100 N. One of the charges is increased by 10% and other is reduced by 10%. Find the new force of repulsion at the same distance.
- Three equal and similar charges are placed at $(-a, 0, 0)$, $(0, 0, 0)$ and $(+a, 0, 0)$. What is the nature of equilibrium of the charge at the origin-

(A) Stable when moved along the Y-axis

(B) Stable when moved along Z-axis

(C) Stable when moved along X-axis

(D) Unstable in all of the above cases
- Two identical simple pendulums A and B, are suspended from the same point. The bobs are given positive charges, with A having more charge than B. They diverge and reach equilibrium, with A and B making angles α_1 and α_2 with the vertical respectively. Which of the two angles is greater.

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ELECTROSTATICS -SHEET: 2(Lecture – 2)

ANSWER KEY

1. $81 \times 8.86 \times 10^{-12}$
2. F , does not change
- 3.

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}^2} \hat{r}_{12}$$

$$\vec{r}_{12} = (1-0)\hat{i} + (1-0)\hat{j} + (1-0)\hat{k} = \hat{i} + \hat{j} + \hat{k}$$

$$|\vec{r}_{12}| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\hat{r}_{12} = \frac{\vec{r}_{12}}{|\vec{r}_{12}|} = \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$$

$$\vec{F}_{21} = \frac{9 \times 10^9 \times \sqrt{3} \times 10^{-19} \times (-10^{-6})}{3} \cdot \frac{(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}}$$

$$= -3 \times 10^{-16} (\hat{i} + \hat{j} + \hat{k}) \text{ Newton}$$

$$4. \quad \hat{u} = -\frac{1}{\sqrt{5}}(\hat{i} + 2\hat{j})$$

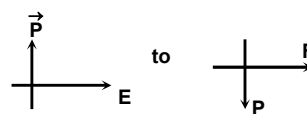
$$4. \quad \vec{F}_1 = \frac{k2q^2}{9}(-\hat{j}) \quad (1)$$

$$\vec{F}_2 = -\frac{kq_2}{9}, (-\hat{i}) \quad (2)$$

$$\vec{F}_R = \vec{F}_1 + \vec{F}_2$$

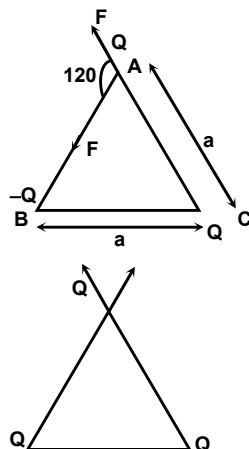
$$= \frac{kq^2}{9}[-\hat{i} - 2\hat{j}]$$

$$\vec{r} = -\hat{i} - 2\hat{j} \Rightarrow \hat{r} = \frac{-\hat{i} - 2\hat{j}}{\sqrt{5}} = -\frac{(\hat{i} + 2\hat{j})}{\sqrt{5}}$$



$$5. \quad (i) \frac{Q^2}{4\pi\epsilon_0 a^2}, (ii) \frac{\sqrt{3}Q^2}{4\pi\epsilon_0 a^2}, (iii) \frac{\sqrt{3}Q^2}{4\pi\epsilon_0 a^2}$$

5.



$$F = \frac{kQ^2}{a^2}$$

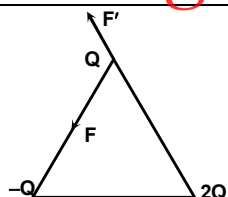
$$F_R = \sqrt{F^2 + F^2 + 2F^2 \cos 120}$$

$$F_R = F = \frac{kQ^2}{a^2} = \frac{Q^2}{4\pi\epsilon_0 a^2}$$

$$F_R = \sqrt{F^2 + F^2 + 2F^2 \cos}$$

$$= \sqrt{3}F = \sqrt{3} \frac{kQ^2}{a^2} = \frac{\sqrt{3}Q^2}{4\pi\epsilon_0 a^2}$$

$$F^1 = \frac{k2Q^2}{a^2}, F = \frac{kQ^2}{a^2}$$



$$F_R = \sqrt{F^2 + F'^2 + 2FF' \cos 120^\circ}$$

$$= \frac{kQ^2}{a^2} \sqrt{1 + 4 - 2}$$

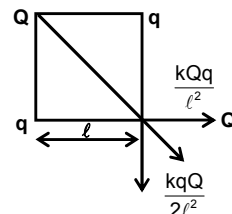
$$= \frac{kQ^2}{a^2} \sqrt{3} = \frac{\sqrt{3}Q^2}{4\pi\epsilon_0 a^2}$$

6. A

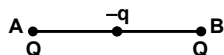
7. $q = -\frac{Q}{2\sqrt{2}}$

7. $\frac{kQ^2}{q\ell^2} + \frac{\sqrt{2}kQ}{\ell^2} = 0$

$$\frac{q}{Q} = -\frac{1}{2\sqrt{2}} \Rightarrow q = -\frac{Q}{2\sqrt{2}}$$



8. C

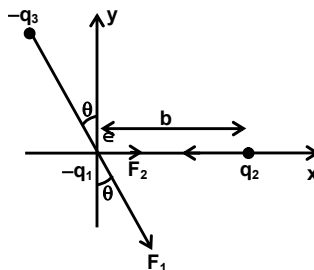


9. $F_1 = \frac{kq_1q_3}{a^2} \quad (1)$

$$F_2 = \frac{kq_1q_2}{b^2} \quad (2)$$

$$F_x = F_1 \sin \theta + F_2$$

$$F_x = \frac{kq_1q_3}{a^2} \sin \theta + \frac{kq_1q_2}{b^2}$$

**LEVEL - II**

1. Let the concerned charge be at C then charge at C will experience the force due to charges at A, B and D. Let these forces respectively be $\vec{F}_A$, $\vec{F}_B$ and $\vec{F}_D$ thus forces are given as

$$\vec{F}_A = \frac{1}{4\pi\epsilon_0} \frac{q^2}{AC^2} \text{ along AC} = \frac{q^2}{4\pi\epsilon_0 2a^2} \left(\frac{\hat{i}}{\sqrt{2}} - \frac{\hat{j}}{\sqrt{2}} \right)$$

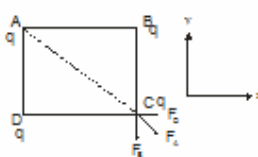
$$\vec{F}_B = \frac{1}{4\pi\epsilon_0} \frac{q^2}{BC^2} \text{ along BC} = \frac{q^2}{4\pi\epsilon_0 a^2} (-\hat{j})$$

$$\vec{F}_D = \frac{1}{4\pi\epsilon_0} \frac{q^2}{DC^2} \text{ along DC} = \frac{q^2}{4\pi\epsilon_0 a^2} (\hat{i})$$

$$\vec{F}_{\text{net}} = \vec{F}_A + \vec{F}_B + \vec{F}_D$$

$$= \frac{q^2}{4\pi\epsilon_0 a^2} \left[\hat{i} \left(\frac{1}{2\sqrt{2}} + 1 \right) - \hat{j} \left(\frac{1}{2\sqrt{2}} + 1 \right) \right]$$

$$|\vec{F}_{\text{net}}| = \sqrt{2} \left(\frac{1}{2\sqrt{2}} + 1 \right) \frac{q^2}{4\pi\epsilon_0 a^2} = \left(\frac{1}{2} + \sqrt{2} \right) \frac{q^2}{4\pi\epsilon_0 a^2}$$



2. Suppose, the same charge 'q' was placed at sixth corner also Then

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 + \vec{F}_6 = 0$$

(Note that resultant is zero due to symmetry of hexagon. This is applicable for any REGULAR geometry)

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5 = -\vec{F}_6$$

$$= -\frac{qQ}{4\pi\epsilon_0 d^2}$$

and direction of force will be opposite to $\vec{F}_6$

3. a circle of radius $r = a/\sqrt{2}$

$$3. \quad F = \frac{k \cdot \theta^2}{(a^2 + x^2)}$$

$$F_y = qF \cos \theta = \frac{2 \cdot k\theta q}{(a^2 + xy)} \times \frac{x}{\sqrt{a^2 + x^2}}$$

$$F_y = 2k\theta q \frac{x}{(a^2 + x^2)^{3/2}}$$

$$\frac{dF_y}{dx} = 2k\theta q \left[\frac{(a^2 + x^2)^{3/2} - \frac{3}{2}(a^2 + x^2)^{1/2}x^2}{(a^2 + x^2)^3} \right] = 0$$

$$= (a^2 + xy^{1/2})[a^2 + x^2 - 3x^2] = 0$$

$$x = \frac{+a}{-\sqrt{2}}$$

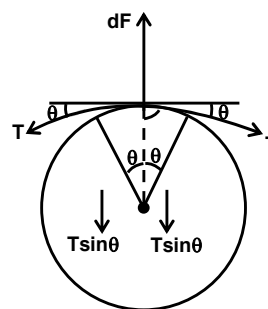
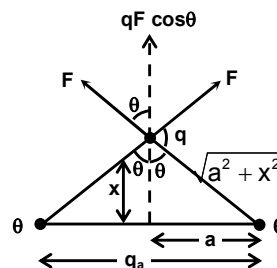
$$4. \quad \Delta T = \frac{qq_0}{8\pi^2 \epsilon_0 r^2}$$

$$4. \quad 2T \sin \theta = dF$$

$$2T \cdot \theta = \frac{kq_0}{R^2} \cdot dq$$

$$2T\theta = \frac{kq_0}{R^2} \cdot \frac{q}{2\pi R} \cdot q\theta R$$

$$T = \frac{kqq_0}{2\pi R^2} = \frac{qq_0}{8\pi \epsilon_0 R^2}$$



5. B, D

$$5. \quad F_1 = F_2$$

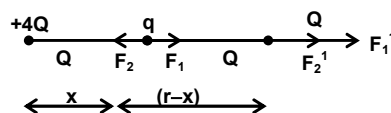
$$\frac{k4\theta q}{x^2} = \frac{k\theta q}{(r-x)^2}$$

$$\frac{2}{x} = \frac{1}{r-x} \Rightarrow 2r - qx = x$$

$$x = \frac{2r}{3}$$

$$4Q$$

$$\frac{r}{3} \text{ from } Q$$



$$\text{also } \vec{F}_1 + \vec{F}_2 = 0$$

$$\frac{k4QQ}{r^2} + \frac{kQq}{(r-x)^2} = 0$$

$$\frac{rQ}{r^2} = -\frac{q}{(r-x)^2}$$

$$q = -4Q \left(\frac{r-x}{r} \right)^2 = -4Q$$

$$q = -\frac{4Q}{9}$$

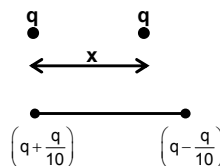
6. 99 N

$$F = \frac{kq^2}{r^2} = 100 \text{ N} \quad (1)$$

$$F' = \frac{k\left(q + \frac{q}{10}\right)\left(q - \frac{q}{10}\right)}{r^2}$$

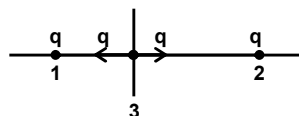
$$F' = \frac{k}{r^2} \left[q^2 - \frac{q^2}{100} \right]$$

$$= 100 - \frac{100}{100} = 99$$



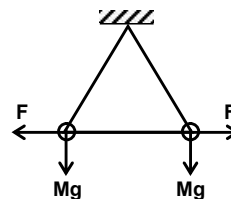
7. C

7. At origin $F_1 = F_2$ but if charge of O is slightly displaced toward right F_2 increases & F_1 decreases, similarly when charge at O is slightly displaced to wards left F_1 increases & F_2 decreases.



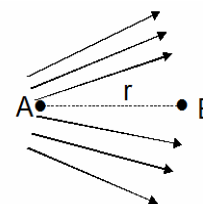
8. Both are equal

8. Though the charges are same force of repulsion will be also same.

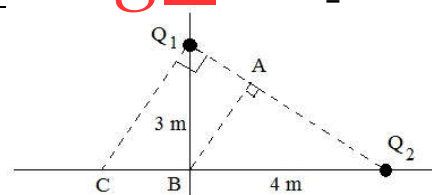


CPP**ELECTROSTATICS -SHEET: 3(Lecture – 3)****Level-I**

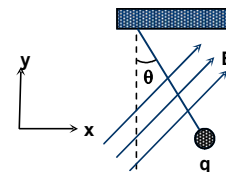
1. A pendulum bob of mass 80mg and carrying a charge of 2×10^{-8} C is at rest in a horizontal uniform electric field of $20,000 \text{ Vm}^{-1}$. Find the tension in the thread of pendulum.
2. Two charges $4q$ and q are placed 30 cm. apart. At what point the value of electric field will be zero.
3. If $Q=2\text{C}$ and force on it is $F=100$ newtons, find the value of field intensity at the position of charge Q .
4. Four equal but like charge are placed at four corners of a square. The electric field intensity at the center of the square due to any one charge is E , then the resultant electric field intensity at centre of square will be
5. Two charges $9e$ and $3e$ are placed at a distance r . The distance of the point where the electric field intensity will be zero is:
6. An electron moves with velocity in x -direction. An electric field acts on it in y -direction. Find the direction of force on the electron.
7. If mass of the electron = 9.1×10^{-31} Kg. Charge on the electron = 1.6×10^{-19} coulomb and $g = 9.8 \text{ m/s}^2$. Find the intensity of the electric field required to balance the weight of an electron.
8. Four charges $+q, +q, -q$ and $-q$ are placed respectively at the corners A, B, C and D of a square of side a , arranged in the given order. Calculate the intensity at O, the centre of the square
9. The tangent drawn at a point on a line of electric force shows the of electric field.
10. Which of the following statements concerning the electrostatics is/are correct(s)-
 (A) Electric lines of force never intersect each other
 (B) Electric lines of force usually start from positive charge and end at the negative charge
 (C) In a charge-free region, electric field lines can be taken to be continuous curves without any breaks
 (D) Electrostatic electric lines of force can never be closed loops
11. If the electric field is uniform, then the electric lines of forces are(Divergent/Convergent/Circular/Parallel)
12. Electric lines of forces-
 (A) Exist everywhere
 (B) Are imaginary
 (C) Exist only in the immediate vicinity of electric charges
 (D) None of the above
13. In fig. shown the electric lines of force emerging from a charged body. If the electric fields at A and B are E_A and E_B are respectively. If the distance between A and B is r then -
 (A) $E_A > E_B$
 (B) $E_A < E_B$
 (C) $E_A = E_B$
 (D) $E_A = (E_B)/r^2$



14. Find the electric field at points A, B and C due to system of charges shown in the figure. Take $Q_1 = 100 \mu\text{C}$ and $Q_2 = 200 \mu\text{C}$



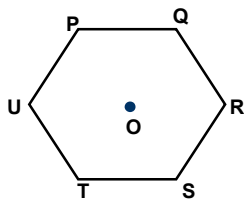
15. A charged cork ball of mass m is suspended on a light string in the presence of a uniform electric field as shown in the adjacent figure. When $\vec{E} = (E_x \hat{i} + E_y \hat{j})$ N/C, the ball is in equilibrium at the angle θ . Find
(a) the charge on the ball, and (b) the tension in the string.



16. Will a positive charge follow electric lines of force if free to move?
17. A particle of mass $6 \mu\text{g}$ carrying a charge of 10^{-9} C, is placed in the electric field of strength $E = 6 \times 10^5$ V/m, find the acceleration acquired by the particle.

Level-II

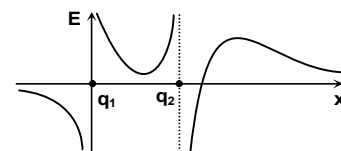
- Three identical charges each of $1 \mu\text{C}$ are kept on the circumference of a circle of radius 1 metre forming equilateral triangle. Find the electric intensity at the center of the circle in N/C.
- Two point charges Q and $-3Q$ are placed certain distance apart. If the electric field at the location of Q be $\vec{E}$ due to $3Q$, then find field at the location of $-3Q$ due to Q .
- An electron of mass m_e , initially at rest, moves through a certain distance in a uniform electric field in time t_1 . A proton of mass m_p , also, initially at rest, takes time t_2 to move through an equal distance in this uniform electric field. Neglecting the effect of gravity, the ratio t_2/t_1 is nearly equal to
(A) 1 (B) $(m_p/m_e)^{1/2}$ (C) $(m_e/m_p)^{1/2}$ (D) 1836
- Six charges of equal magnitude are placed at six corners of a regular hexagon. Find arrangement the charges in order PQRSTU which produce double electric field at centre as compared to electric field produce by single charges $+q$ at R



- (A) +++- - - (B) + - + - + -
(C) - + + - + - (D) - + + + - -
5. Under the influence of the Coulomb field of charge $+Q$, a charge $-q$ is moving around it in an elliptical orbital. Find out the correct statement(s) -
(A) The angular momentum about Q of the charge $-q$ is constant
(B) The linear momentum about Q of the charge $-q$ is constant
(C) The angular velocity of the charge $-q$ is constant
(D) The linear speed of the charge $-q$ is constant

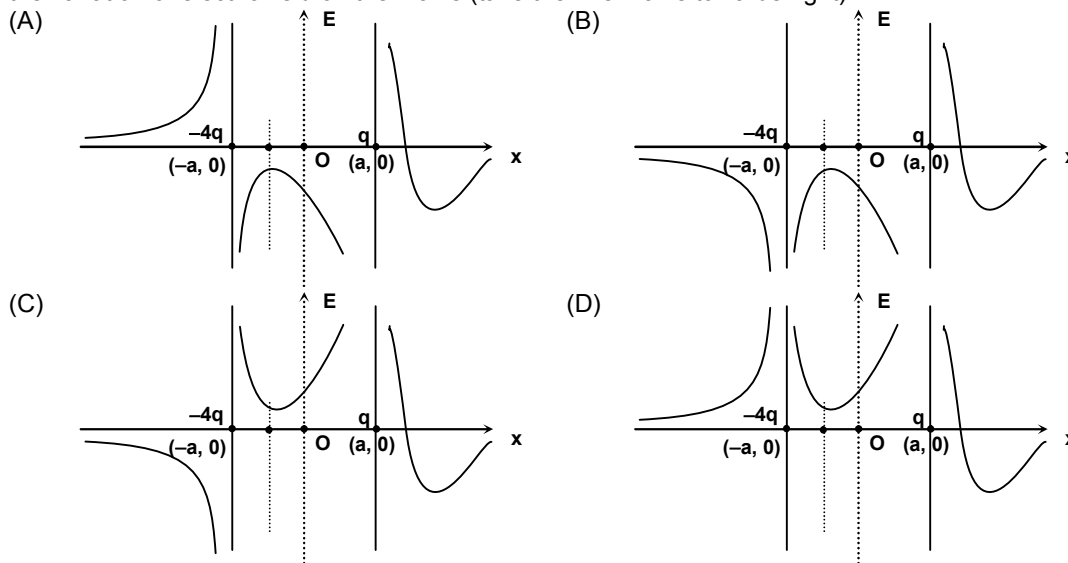
6. Electric field in a region is given as $\vec{E} = (10 - 5x)\text{N/C} \hat{i}$. A charge particle of mass 5 kg and charge $Q (= 1 \text{ C})$ is situated at origin and free to move in given electric field. Then choose the correct options.
- (A) Motion of charge particle is Oscillatory
 (B) Maximum displacement of charge particle from origin is 4 SI units
 (C) Maximum velocity gain by charge particle is 2 SI units
 (D) The position of charge particle, when velocity gained by particle is maximum, is 2 SI units

7. Two charges q_1 and q_2 are kept on x-axis and the variation of electric field strength at different points on x-axis is described in the adjacent figure graphically. Choose correct statement about nature and magnitude of q_1 and q_2 .

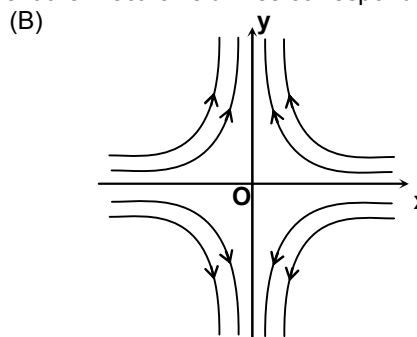
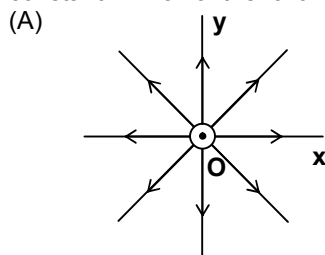


- (A) $q_1 +ve, q_2 +ve, |q_1| > |q_2|$
 (B) $q_1 +ve, q_2 -ve, |q_1| > |q_2|$
 (C) $q_1 +ve, q_2 -ve, |q_1| < |q_2|$
 (D) $q_1 -ve, q_2 +ve, |q_1| < |q_2|$

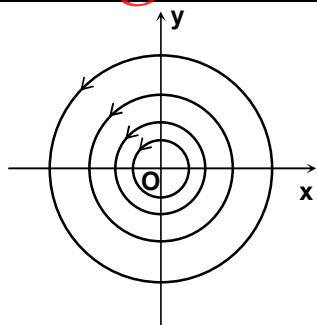
8. Two charge particle $-4q$ and q lies on x-axis as shown in the figure. Which of the following graphs will show the variation of electric field on the x-axis (take the +ve x-axis towards right).



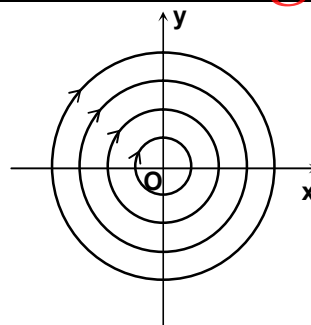
9. Electric field at point P in a region whose coordinates are (x, y, z) is given by : $\vec{E} = E_0 \frac{(-y\hat{i} + x\hat{j})}{x^2 + y^2}$ where E_0 is constant. Which of the following can best represent the Electric field lines corresponding to this field?



(C)



(D)



CPP

ELECTROSTATICS -SHEET: 3(Lecture – 3)

ANSWER KEY

Level-I

1. $8.8 \times 10^{-4} \text{ N}$

1. $M = 80 \text{ mg}$, $q = 2 \times 10^{-8} \text{ C}$, $E = 20,000 \text{ v/m}$ $M = 8 \times 10^{-5} \text{ kg}$

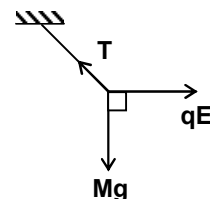
$$T = \sqrt{(qE)^2 + (mg)^2}$$

$$= \sqrt{(2 \times 10^{-8} \times 2 \times 10^4)^2 + (8 \times 10^{-5} \times 10)^2}$$

$$= \sqrt{16 \times 10^{-8}} = 4\sqrt{5} \times 10^{-4}$$

$$= \sqrt{80 \times 10^{-8}} = 4\sqrt{5} \times 10^{-4}$$

$$\Rightarrow 4 \times 2.24 \times 10^{-4} \text{ N.}$$



2. 10 cm. away from q and between the charge

2. Let assume the E.F will be zero at Pt. P.

$$E_A \text{ at Pt. P} = E_B \text{ at Pt P}$$

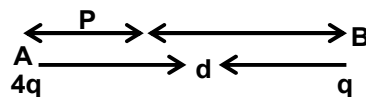
$$\frac{K(4q)}{x^2} = \frac{Kq}{(30-x)^2}$$

$$\Rightarrow \left(\frac{30-x}{x}\right)^2 = \left(\frac{1}{2}\right)^2$$

$$\Rightarrow \left(\frac{30}{x} - 1\right)^2 = \frac{1}{2} \quad \text{T distance of 20 cm from charge } 4q.$$

$$\frac{30}{x} = \frac{3}{2}$$

$$x = 20 \text{ cm}$$

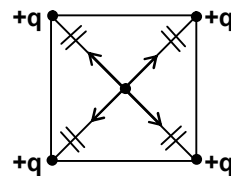


3. 50 N/C

3. $E = \frac{F}{Q} = \frac{100}{2} = 50 \text{ N/C}$

4. Zero

4. Since the charges are like and equal in magnitude so, the resultant will be zero at centre.



5. $\left(\frac{r}{1 + \sqrt{1/3}} \right)$ from 9e charge

5. If the test charge is placed b/q the two charges E_p (due to $3q$)
 $= \frac{3kq}{x^2}$

$$E_p \text{ (due to } q) = \frac{kq}{(r-x)^2}$$

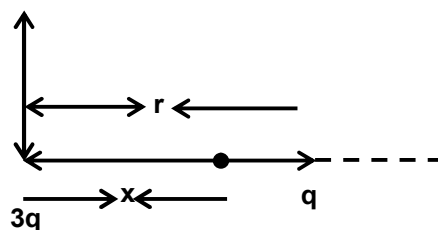
At pt P E.F will be equal and opposite by both the charges :-

$$\frac{3kq}{x^2} = \frac{kq}{(r-x)^2} \Rightarrow \left(\frac{r-x}{x} \right)^2 = \frac{1}{3}$$

$$\frac{r}{x} - 1 = \pm \frac{1}{\sqrt{3}} \Rightarrow \frac{r}{x} = \frac{1}{\sqrt{3}} + 1$$

$$X = \frac{r}{\left(1 + \frac{1}{\sqrt{3}} \right)}$$

$$x = \frac{r}{1 + \frac{1}{\sqrt{3}}}$$



6. -ve direction of Y-axis
 6. Electric field is along (+ve) y-axis. Charge on electron $q : -e$
 $F_e = q\vec{E} = -e\vec{E}$
 So, the direction of force on electron is along -ve y - axis.

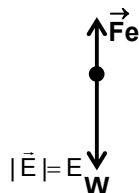
7. $5.6 \times 10^{-11} \text{ N/C}$

7. $W = F_e$

$$\Rightarrow mg = qE$$

$$\Rightarrow E \cdot \frac{mg}{q} = \frac{9.1 \times 10^{-31} \times 98}{16 \times 10^{-19}}$$

$$E = 5.6 \times 10^{-11} \text{ N/C}$$



8. $\frac{4\pi\epsilon_0 \cdot a^2}{4\sqrt{2}q}$

8. Magnitude of E.F by all the four charges at centre is

$$|\vec{E}_e| \text{ (at 0)} = \frac{kq^2}{\left(\frac{1}{\sqrt{2}}\right)^2} = \frac{2kq^2}{a^2} = E$$

$$\vec{E}_A \text{ (at 0)} = E \text{ along } \overline{DC}$$

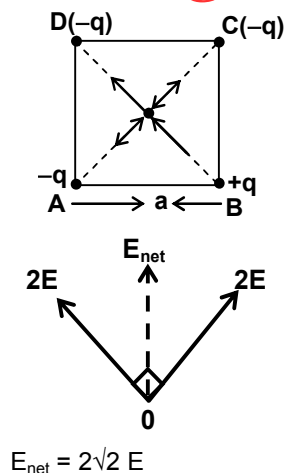
$$\vec{E}_B \text{ (at 0)} = E \text{ along } \overline{OD}$$

$$\vec{E}_C \text{ (at 0)} = E \text{ along } \overline{DC}$$

$$\vec{E}_D \text{ (at 0)} = E \text{ along } \overline{OD}$$

$$|\vec{E}_0| = 2\sqrt{2}E = 2\sqrt{2} \times \frac{2 \times 1 \times q}{4\pi\epsilon_0 \times 2}$$

$$= \frac{\sqrt{2}q}{\pi\epsilon_0 a^2}$$



9. Direction

10. ABCD

11. Parallel

11. Parallel (e.g field lines is parallel due to a infinite sheet as E.F is uniform due to this)

12. B

13. A

13. (magnitude of E.F depends upon the density of field lines larger the density greatest the E.F and vice – versa)

14. $EA = 1.02 \times 10^5 N$ and $EB = 3.25 \times 10^5 N$. Calculate EC your electric lines of force.

14. $q_1 = 100 \mu C$ (q)

$$q_2 = 200 \mu C$$

In $\triangle OBA$

$$\sin \theta = \frac{AB}{OB}, \quad \frac{AB}{3m} \quad (i)$$

$$\text{In } \triangle OBD, \sin \theta = \frac{BD}{OD} = \frac{4m}{5m} \quad (ii)$$

From (i), (ii) :-

$$\frac{AB}{3m} = \frac{4}{5} \Rightarrow AB = \frac{12}{5}m$$

$$OA^2 = OB^2 - AB^2 = 9 - \frac{144}{25} = \frac{81}{25}$$

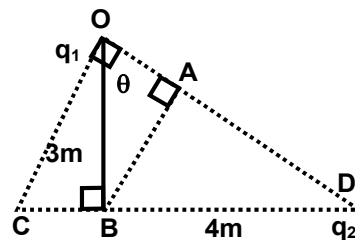
$$OA = \frac{9}{5}m$$

$\triangle DAB \approx \triangle DOC$

$$\Rightarrow \frac{AD}{OD} = \frac{AB}{OC} \Rightarrow \frac{5 - \frac{9}{5}}{5} = \frac{12}{5OC}$$

$$\Rightarrow \frac{16}{25} = \frac{12}{5OC} \Rightarrow OC = \frac{15}{4}m.$$

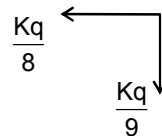
$$E_A \text{ (due to } q_1) = \frac{Kq}{OA^2} = \frac{25Kq}{81}$$



$$E_A \text{ (due to } q_2) = \frac{K \times 2q}{AD^2} = \frac{50Kq}{256}$$

$$E_B \text{ (due to } q_1) = \frac{Kq}{9}$$

$$E_B \text{ (due to } q_2) = \frac{2Kq}{16} = \frac{Kq}{8}$$



15. For equilibrium of ball

$$\sum \vec{F} = 0$$

$$\therefore T \cos \theta \hat{j} - T \sin \theta \hat{i} + qE_x \hat{i} + qE_y \hat{j} - mg \hat{j} = \vec{0}$$

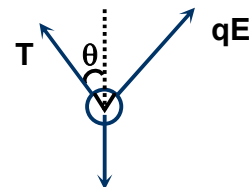
$$\Rightarrow T \sin \theta = qE_x \quad \dots(1)$$

$$\text{and } T \cos \theta = mg - qE_y \quad \dots(2)$$

From (1) and (2),

$$q = \frac{mg \tan \theta}{E_x + E_y \tan \theta}$$

$$T = \frac{mg E_x \sec \theta}{E_x + E_y \tan \theta}$$



16. No, if ELECTRIC LINES OF FORCE are curved, direction of velocity(motion) and acceleration(force) will be different.

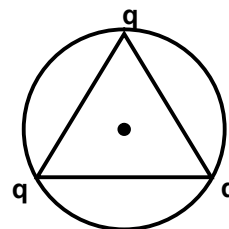
17. 105 m/sec^2

$$17. |\vec{a}| = \frac{|\vec{F}_e|}{m} = \frac{qE}{m} = \frac{10^{-9} \times 6 \times 10^5}{6 \times 10^{-6} \times 10^{-3}} = 10^5 \text{ m/s}^2$$

Level-II

1. Zero

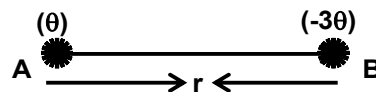
1. The centered of the circle and centered of the triangle will coincide of the same point. Hence the intensity will be zero at centre (at centered of equilateral triangle intensity is zero)



2. $\vec{E}/3$

$$2. |\vec{E}_A| \text{ (due to } +3\theta) = \frac{k \times 3\theta}{r^2} = \frac{3k\theta}{r^2} = E$$

$$|\vec{E}_B| \text{ (due to } -\theta) = \frac{k\theta}{r^2} = \frac{E}{3}$$



3. B

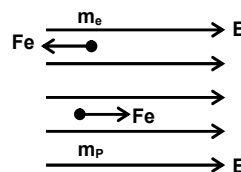
3. Electrostatic force will be equal is magnitude in both electron and proton as the carry same charge.

$$A_e(m_e) = F_e = \frac{qE}{m_e}$$

(acceleration of electron mass m_e)

$$a_p(m_p) = \frac{F_p}{m_p} = \frac{qE}{m_p}$$

$$t_1 = \sqrt{2a_e s} = \sqrt{\frac{2qEs}{m_e}}$$

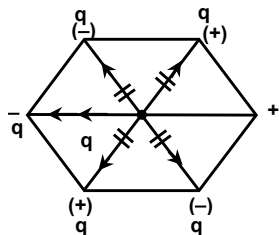


$$t_2 = \sqrt{2a_p s} = \sqrt{\frac{2qEs}{m_e}}$$

$$\frac{t_1}{t_2} = \sqrt{\frac{m_p}{m_e}} = \left(\frac{m_p}{m_e}\right)^{1/2}$$

4. B

4.



5. A

5. Angular momentum is conserved due to a central force.

6. ABCD

$$\vec{F}_e = q\vec{E} = (10 - 5x)\text{N/C} \hat{i}$$

$$Ma = (10 - 5x)$$

$$= \frac{10 - 5x}{5} = 2x$$

Acceleration is directing proportional to the negative of displacement hence the particle will perform oscillatory motion. When

$$a = 0 \Rightarrow 2 - x = 0 \Rightarrow x = 2 \text{ mean position will be at } x = 2 \text{ units} \quad v_{\max} = \sqrt{A^2 - x^2} = \sqrt{2^2 - 0^2} = 2 \text{ unit (at mean position)}$$

$$A = 4 \text{ unit.}$$

7. B

7. q_1 will be +ve and q_2 will be negative as the E.F. lines are in 1st and 3rd quadrant due to (+ve) charge and 2nd and 4th quadrant due to (-ve) charge. And $|q_1| > |q_2|$ B.C. field lines of q_1 dominate on q_2 as shown in figure.

8. A

9. C

Electric field is tangential direction at all points ($\vec{E} \cdot \hat{r} = 0$) and decreasing with increase in r .
(r = magnitude of radius vector of a point)

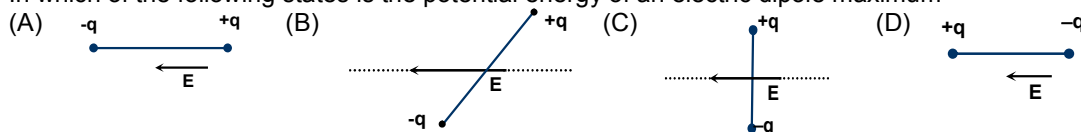
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ELECTROSTATICS -SHEET: 4(Lecture – 4)

LEVEL – I

1. A system has two charges $q_A = 2.5 \times 10^{-7} \text{ C}$ and $q_B = -2.5 \times 10^{-7} \text{ C}$ located at points A: (0, 0, -15 cm) and B: (0, 0, +15 cm), respectively. What are the total charge and electric dipole moment of the system?
2. Electric charges q , q , $-2q$ are placed at the corners of an equilateral triangle ABC of side l . Find the magnitude of electric dipole moment of the system
3. Find the electric intensity due to a dipole of length 10 cm and having a charge of $500 \mu\text{C}$, at a point on the axis at a distance 20 cm from one of the charges in air.
4. What is the angle between the electric dipole moment and the electric field strength due to it on the equatorial line.
5. Two electric dipoles of moment P and $64 P$ are placed in opposite direction on a line at a distance of 25 cm. The electric field will be zero at point between the dipoles find the distance from the dipole of moment P .
6. What is the ratio of the electric field due to an electric dipole on its axis and on the perpendicular bisector of the dipole (assuming large distances).

7. In which of the following states is the potential energy of an electric dipole maximum

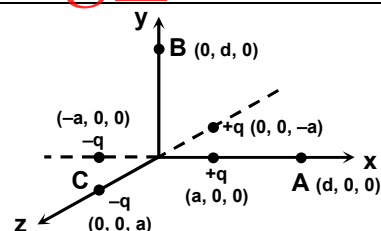


8. When an electric dipole $\vec{P}$ is placed in a uniform electric field $\vec{E}$ then at what angle between $\vec{P}$ and $\vec{E}$ the value of torque will be maximum.
9. An electric dipole consists of two opposite charges each of magnitude $1 \times 10^{-6} \text{ C}$ separated by a distance 2cm. The dipole is placed in an external field of $10 \times 10^5 \text{ N/C}$. Find the maximum torque on the dipole.
10. An electric dipole has the magnitude of its charge as q and its dipole moment is p . It is placed in a uniform electric field E . If its dipole moment is along the direction of the field, the force on it and its potential energy are respectively
 (A) $2q \cdot E$ and minimum
 (B) $q \cdot E$ and $p \cdot E$
 (C) Zero and minimum
 (D) $q \cdot E$ and maximum
11. An electric dipole of moment p is placed normal to the lines of force of electric intensity E , find the work done in deflecting it through an angle of 180° .
12. Two opposite and equal charges $4 \times 10^{-8} \text{ coulomb}$ when placed $2 \times 10^{-2} \text{ cm}$ away, form a dipole. If this dipole is placed in an external electric field $4 \times 10^8 \text{ newton/coulomb}$, find the value of maximum torque and the work done in rotating it through 180° .

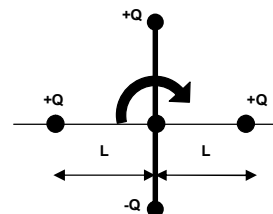
LEVEL – II

1. An electric dipole is placed along the x-axis at the origin O. A point P is at a distance of 20 cm from this origin such that OP makes an angle $\frac{p}{3}$ with the x-axis. If the electric field at P makes an angle θ with the x-axis, find the value of θ .

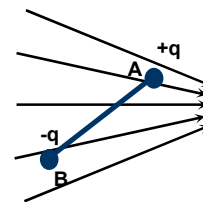
2. Four charges are fixed on the x-z plane as shown in the figure. Find the work done by the electric force applied by this system of charges on a positive charge particle $+Q$, to displace it from point $A(d, 0, 0)$ to point $B(0, d, 0)$ if $d \gg a$.



3. In the diagram shown, the $+Q$ and $-Q$ charges are connected by a non-conducting rod, which can freely rotate about the fixed axis passing through the centre. The other two $+Q$ charges are fixed at the positions shown. Determine nature of equilibrium of dipole.

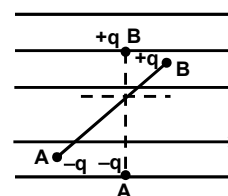


4. Two connected charges of $+q$ and $-q$ respectively are at fixed distance AB apart in a non-uniform electric field whose lines of force are shown in the figure. The resultant effect on the two charges is
 (A) a torque vector in the plane of the paper and no resultant force.
 (B) a resultant force in the plane of the paper and no torque.
 (C) a torque vector normal to the plane of the paper and no resultant force.
 (D) a torque vector normal to the plane of the paper and a resultant force in the plane of the paper.



5. A dipole is placed at a distance $2s$ and s respectively from charges $+4Q$ and $-Q$. The dipole is inclined at an angle θ from the line joining the two charges. If F_1 and F_2 are the forces acting on the positive and negative charges of the dipole, then: (Positive charge of dipole faces charge $+4Q$)
 (a) $\vec{F}_1 > \vec{F}_2$ (b) $\vec{F}_1 < \vec{F}_2$ (c) $\vec{F}_1 = \vec{F}_2$ (d) $\vec{F}_1 + \vec{F}_2 = 0$

6. A dipole consists of a light rod of length ℓ , having two point charges $+q$ and $-q$ of same mass m joined at ends A and B of the rod. A uniform electric field E is present in the region and perpendicular to the length of the rod. The rod is released from rest. Find the angular velocity of the rod, when it becomes parallel to the electric field for the first time.



CPP

ELECTROSTATICS -SHEET: 4(Lecture – 4)

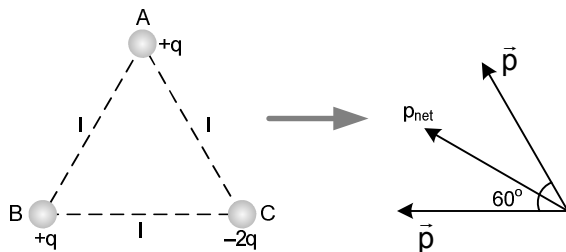
ANSWER KEY

LEVEL - I

1. Total charge is zero. Dipole moment = 7.5×10^{-8} C m along z-axis.

2. $\sqrt{3}gl$

2.



$$p_{\text{net}} = \sqrt{p^2 + p^2 + 2pp \cos 60^\circ} = \sqrt{3}p = \sqrt{3}ql \quad (\because p = ql)$$

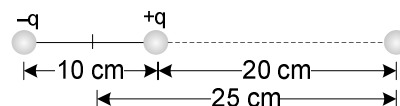
3. $6.25 \times 10^7 \text{ N/C}$

3. By using $E = 9 \times 10^9 \times \frac{2pr}{(r^2 - l^2)^2}$; where

$$p = (500 \times 10^{-6}) \times (10 \times 10^{-2}) = 5 \times 10^{-5} \text{ C m},$$

$$r = 25 \text{ cm} = 0.25 \text{ m}, l = 5 \text{ cm} = 0.05 \text{ m}$$

$$E = \frac{9 \times 10^9 \times 2 \times 5 \times 10^{-5} \times 0.25}{\{(0.25)^2 - (0.05)^2\}^2} = 6.25 \times 10^7 \text{ N/C}$$



4. 180°

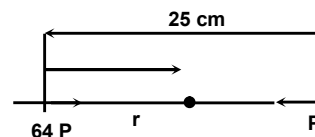
4. The direction of electric field intensity at a point on the equatorial line of the dipole is opposite to the direction of dipole moment.

5. 5 cm

$$\frac{k64P}{r^3} = \frac{kp}{(25-r)^3} = \left(\frac{r}{25-r}\right)^3 = 64 \frac{r}{25-r} = 4$$

$$5r = 100$$

$$R = 20 \text{ cm}$$



6. 2 : 1

6. $E\text{-axis} = \frac{2kP}{r^3}$

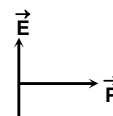
$$\frac{\text{Equation water}}{r^3} = \frac{k8}{r^3} \quad \frac{E\text{axis}}{\text{Equation}} = \frac{2}{1}$$

7. A

7. $U = -\vec{P} \cdot \vec{E}$ U is max is A

8. 90°

8. $T = \vec{P} \times \vec{E}$ max if $\theta = 90^\circ$ or 270°



9. Zero

9. $T_{\text{max}} = PE = (10^{-6} \times 2 \times 10^{-2}) 10 \times 10^5 = 2 \times 10^{-2} \text{ N-m}$

10. C

11. $2 \times 10^{-3} \text{ N-m}$

11. $W = 0$

$$U = - \vec{P} \cdot \vec{E}$$

$$U_i = 0$$

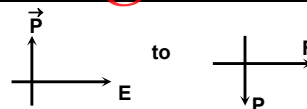
$$U_f = 0$$

12. $32 \times 10^{-4} \text{ Nm and } 64 \times 10^{-4} \text{ J}$

12. $\zeta_{\max} = PE = (4 \times 10^{-8} \times 2 \times 10^{-4}) 4 \times 10^8$

$$= 32 \times 10^{-4} \text{ N-m}$$

$$W = U_f - U_i = 2PE = 64 \times 10^{-4} \text{ N-m}$$

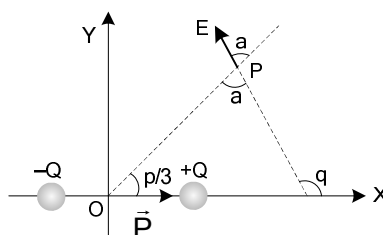
**LEVEL - II**

1. $\frac{p}{3} + \tan^{-1} \frac{\sqrt{3}}{2} \frac{\ddot{\theta}}{\ddot{\theta}}$

1. $q = \frac{p}{3} + a$ where $\tan a = \frac{1}{2} \tan \frac{p}{3}$

$$p \quad a = \tan^{-1} \sqrt{3}/2 \text{ so,}$$

$$q = \frac{p}{3} + \tan^{-1} \sqrt{3}/2$$



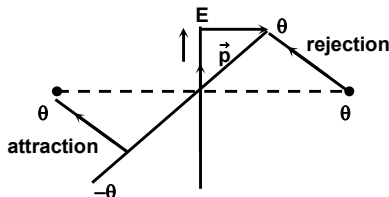
2. $W_{\text{electric}} = U_i - U_f = \frac{(q2a)(Q)}{4\pi\epsilon_0 a^2}$

3. Neutral

3. $\tau = 0$

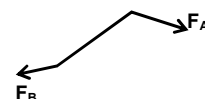
Eva rafter

Displacing

From equation $\Rightarrow$ Neutral equation

4. D

4. Since field at A is stronger (crowding of field lines) & field at B is weaker (less crowding) torque normal to plane of paper & resultant force is plane of paper.

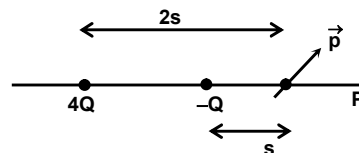


5. D

5. $E_{P=0}$

$$\Rightarrow F_{\text{net}} = 0$$

$$\Rightarrow \vec{F}_1 + \vec{F}_2 = 0$$



6. $\Delta W = -\Delta u$

$$\frac{m\ell^2}{4} \omega^2 = q\ell E$$

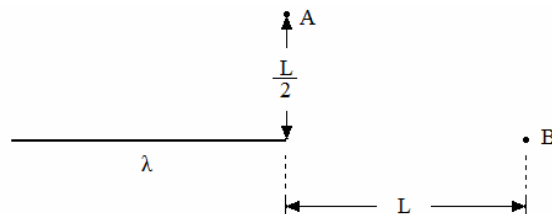
$$\Rightarrow \omega = 2\sqrt{\frac{qE}{m\ell}}$$

CPP

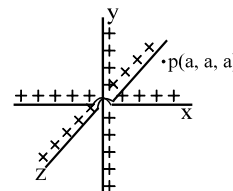
ELECTROSTATICS -SHEET: 5(Lecture – 5)

Level-I

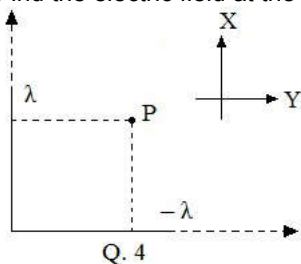
1. Find the electric field intensity due to a uniform linear charge distribution with linear charge density ' λ ' and length ' L ' at a the point A and B shown in the figure.



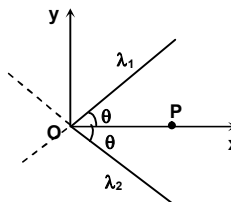
2. Three infinitely long uniformly charged (linear charge density = λ) thin wire are placed along x, y & z axis, then find the electric field strength at point P (a, a, a).



3. Find the electric field at the point P due to two semi-infinite wires shown in the figure.

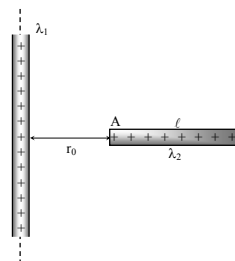


4. Two infinitely long line charges λ_1 and λ_2 are placed symmetric to x-axis as shown in the figure. Match the following for direction of electric field at point P on the x-axis.



Column I		Column II	
(A)	Both λ_1 and λ_2 are positive	(p)	positive x-axis
(B)	Both λ_1 and λ_2 are negative	(q)	positive y-axis
(C)	λ_1 is positive but λ_2 is negative	(r)	negative x-axis
(D)	λ_1 is negative but λ_2 is positive	(s)	negative y-axis

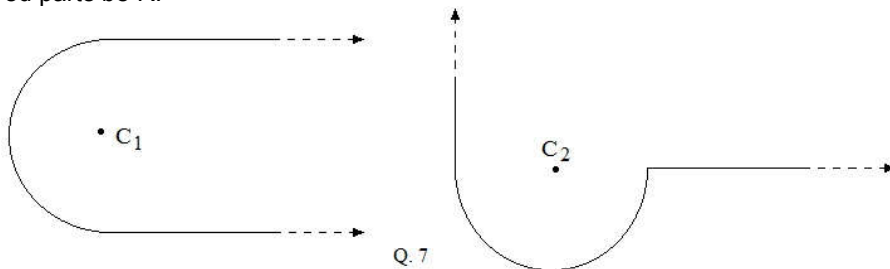
5. An infinitely long string uniformly charged with a linear density λ_1 and a segment of length ℓ uniform charged with linear density λ_2 lie in a plane at right angles to each other and separated by a distance r_0 . Find the force with which these two interact.



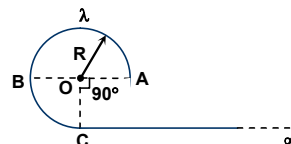
6. A small circular ring has a uniform charge distribution. On a far-off axial point distance x from the centre of the ring, the electric field is proportional to x^{-k} , find k .
7. Find the electric field intensity due to a uniformly charged semicircular ring of radius R and linear charge density ' λ ' at
(a) centre of the ring

(b) a point 'x' distance from the centre of ring on its axis.

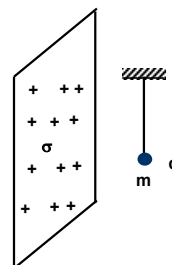
8. Find the maximum value of the electrostatic force on a charge 'q' placed along the axis of the non-conducting circular ring of radius 'a' if charge Q is uniformly distributed on the circumference of the ring.
9. Find the electric field intensity at the centres C_1 and C_2 due to the uniform linear charge density λ with its two ends extending infinitely. The curved part is a semicircle as shown in figure in parts (a) and (b). Radius of curved parts be R.



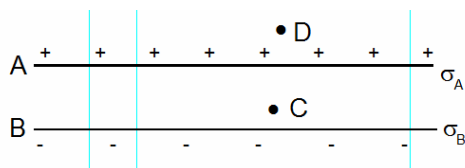
10. A semi infinite line charge of linear charge density λ has the shape as shown in the figure. Portion ABC forms three-fourth of a circle of radius R while the straight portion from C to infinity is parallel to BOA. Find the field at the centre of circle (O).



11. A system consists of a thin charged wire ring of radius R and a very long uniformly charged thread oriented along the axis of the ring, with one of its ends coinciding with the centre of the ring. The total charge on the ring is equal to 'q'. The charge of the wire (per unit length) is equal to λ . Find the interaction force between the ring and the thread.
12. A particle of mass m and charge $-Q$ is constrained to move along the axis of a ring of radius 'a'. The ring carries a uniform charge density $+\lambda$ along its length. Initially, the particle is in the centre of the ring where the force on it is zero. Find the period of oscillation of the particle when it is slightly displaced from its equilibrium position.
13. A small charged particle of mass m and charge q is suspended by an insulated thread in front of a very large sheet of charge density σ . Find the angle made by the thread with the vertical in equilibrium.



14. Find the surface electric charge density on the surface of a circular disc of radius $a = 10$ cm, if a point charge of mass 200 gm and charge $100 \mu\text{C}$ is in equilibrium at a height of 10 cm above the centre of the disc.
15. Two parallel plates of infinite dimensions are uniformly charged. The surface charge density on one is σ_A will on the other is σ_B , field intensity at point C will be-

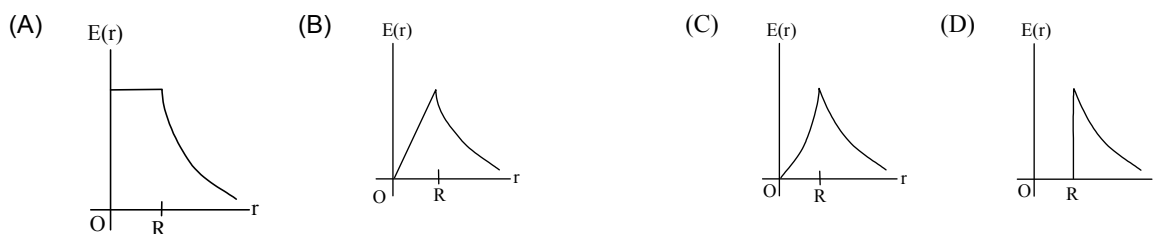


16. Match the electric field lines with the charge configuration.

Column – I		Column – II	
(A)	Three equal positive charges placed at the vertices of an equilateral triangle	(p)	

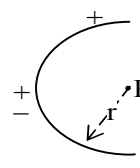
(B)	Thin, circular disc of uniform +ve charge distribution	(q)	
(C)	Two point charges of +q and -2q separated by a small distance	(r)	
(D)	Conductor of finite length with uniform +ve charge distribution	(s)	

17. A thin spherical shell of radius R has charge Q spread uniformly over its surface. Which of the following graphs most closely represents the electric field $E(r)$ produced by the shell in the range $0 \leq r < \infty$, where r is the distance from the centre of the shell?

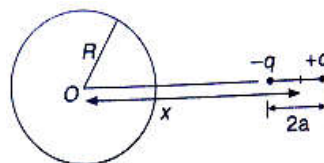


LEVEL – II

1. A thin glass rod is bent into a semicircle of radius r . A charge $+q$ is uniformly distributed along the upper half and a charge $-q$ is uniformly distributed along the lower half. As shown in Figure. Find the electric field $\vec{E}$ at P , the centre of the semicircle.

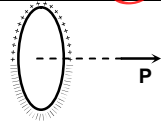
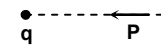


2. An electric dipole is placed at a distance x from centre O on the axis of a charged ring of radius R and charge Q uniformly distributed over it. Find the net force acting on the dipole. ($x \gg a, R$)



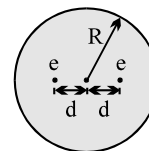
3. A short dipole of dipole moment P is kept near the three different sources of electric field (an infinite line charge, a point charge, a ring of charge) in column I. The force F experienced by the dipole, the torque τ acting on the dipole and the potential energy U of dipole are given in column – II.

Column-I		Column-II	
(A)		(p)	$F = 0$
(B)		(q)	$\tau = 0$

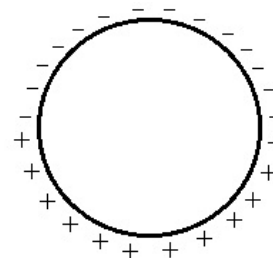
(C)		(r)	$F \neq 0$
(D)		(s)	$U < 0$

4. Let $P(r) = \frac{Q}{\pi R^4} r$ be the charge density distribution for a solid sphere of radius R and total charge Q . For point 'p' inside the sphere at distance r_1 from the centre of the sphere, find the magnitude of electric field is –

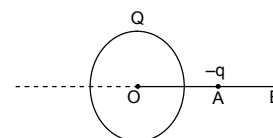
5. Using Thomson's model of the atom, consider an atom consisting of two electrons, each of charge $-e$, embedded in a sphere of charge $+2e$ and radius R . In equilibrium each electron is at distance d from the centre of the atom. What is equilibrium separation between electrons?



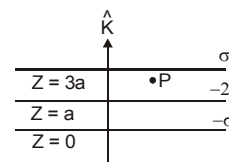
6. Find the electric field intensity at a point P due to a charged rod AB of length L , whose linear charged density varies as a function of distance ' x ' from point A as $\lambda(x) = \lambda_0 x / L$. Given, $\lambda_0 = 2 \mu\text{C/m}$, $L = 50 \text{ cm}$, and point P lies along the line AB at a distance of L from B and $2L$ from A .
7. A large sheet carries uniform surface charge density σ . A rod of length 2ℓ has a linear charge density λ on one half and $-\lambda$ on the other half. The rod is hinged at mid point O (O is on the sheet) and makes θ with the normal to the sheet. Calculate the torque experienced by the rod.
8. Find the electric field at the centre of the non-conducting circular ring of radius ' a ' with charge density ' λ ' and ' $-\lambda$ ' on each semicircular part as shown in figure above.



9. A uniformly charged fixed ring (Q) of radius R has a small charge ($-q$) on its axis at a point A ($OA = d$ and $d \ll R$). When the charge was released the distance AO was covered in time T . Then which of the following statements is/are true.



- (A) For $OB = 2d$ if charge was released from B then the distance AO would be covered in a time $\frac{T}{6}$.
- (B) For $OB = \sqrt{2}d$ if charge was released from B then the distance AO would be covered in a time $\frac{T}{2}$.
- (C) If Q is doubled then T doubles
- (D) If R is doubled then T increases
10. Three large charged sheets having surface charge density as shown in the figure. The sheets are placed parallel to XY plane. Find the electric field at point P .



11. A thin stationary ring of radius 1 m has a positive charge $10 \mu\text{C}$ uniformly distributed over it. A particle of mass 0.9 gm and having a negative charge of $1 \mu\text{C}$ is placed on the axis at a distance of 1 cm from the centre of the ring and released then find the time period of oscillation of particle.

CPP

ELECTROSTATICS -SHEET: 5(Lecture – 5)

ANSWER KEY

LEVEL – I

$$1. \quad (a) \quad \vec{E} = \frac{2k\lambda}{L} \left\{ \left(1 - \frac{1}{\sqrt{5}} \right) \hat{i} + \frac{2}{\sqrt{5}} \hat{j} \right\} \quad (b) \quad E = \frac{k\lambda}{2L}$$

$$1. \quad (a) \quad E_y = \frac{k\lambda}{r} (\sin \alpha + \sin \beta) \quad \sin \alpha = \frac{2L}{\sqrt{5}L} = \frac{2}{\sqrt{5}}$$

$$= \frac{k\lambda}{\left(\frac{L}{2} \right)} \left[\frac{2}{\sqrt{5} + 0} \right]$$

$$\vec{E}_y = \frac{4k\lambda}{\sqrt{5}L} \cdot \hat{j} \quad (1)$$

$$E_x = \frac{k\lambda}{r} (\cos \beta - \cos \alpha)$$

$$= \frac{k\lambda}{(L/2)} \left[1 - \frac{K/2}{\sqrt{5}K/2} \right]$$

$$\vec{E}_x = \frac{2k\lambda}{L} \left[1 - \frac{1}{\sqrt{5}} \right] \hat{i} \quad (2)$$

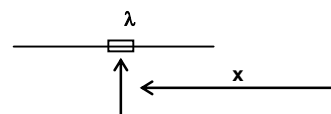
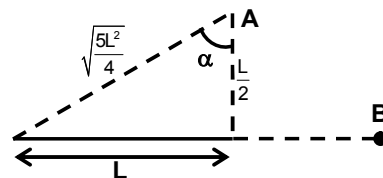
$$\vec{E} = \vec{E}_x + \vec{E}_y$$

$$\vec{E} = \frac{2k\lambda}{L} \left[\left(1 - \frac{1}{\sqrt{5}} \right) \hat{i} + \frac{2}{\sqrt{5}} \hat{j} \right]$$

$$(b) \quad dq = \lambda dx$$

$$dE = \frac{k dq}{x^2} \Rightarrow \int dE = \int \frac{k \cdot \lambda dx}{x^2}$$

$$E = k\lambda \left[-\frac{1}{x} \right]_L^{2L} = -k\lambda \left[\frac{1}{2L} - \frac{1}{L} \right] = \frac{k\lambda}{2L}$$



$$2. \quad \frac{\lambda}{2\pi\epsilon_0 a} [\hat{i} + \hat{j} + \hat{k}]$$

2. Due to wire
Along z axis

$$E_1 = \frac{\lambda}{2\pi\epsilon_0 \sqrt{2}a} [\cos \hat{i} + \cos \hat{j}]$$

$$\vec{E}_1 = \frac{\lambda}{4\pi\epsilon_0 a} (\hat{i} + \hat{j}) \quad (1)$$

Due to wire on x-axis

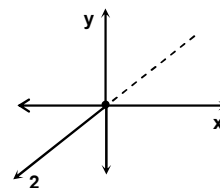
$$\vec{E}_2 = \frac{\lambda}{4\pi\epsilon_0 a} (\hat{j} + \hat{k}) \quad (2)$$

Due to wire on y-axis

$$\vec{E}_3 = \frac{\lambda}{4\pi\epsilon_0 a} (\hat{i} + \hat{k}) \quad (3)$$

$$\vec{E} = \vec{E}_1 + \vec{E}_2 + \vec{E}_3$$

$$= \frac{\lambda}{4\pi\epsilon_0 a} [2\hat{i} + 2\hat{j} + 2\hat{k}]$$



$$\vec{E} = \frac{\lambda}{2\pi\epsilon_0 a} [\hat{i} + \hat{j} + \hat{k}]$$

$$3. \quad \vec{E} = \frac{k\lambda}{x} (1 + \sqrt{2}) (\hat{j} - \hat{i})$$

$$3. \quad E_1 = \frac{k\lambda}{a} \left[1 + \frac{1}{\sqrt{2}} \right]$$

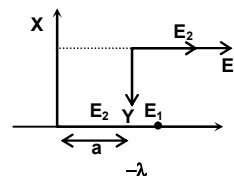
$$E_2 = \frac{k\lambda}{\sqrt{2}a}$$

$$\vec{E}_R = \vec{E}_X + \vec{E}_Y$$

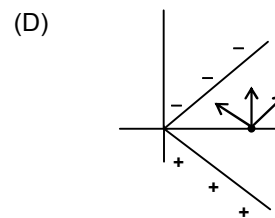
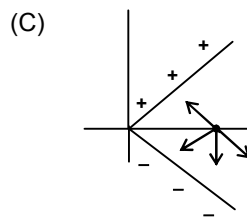
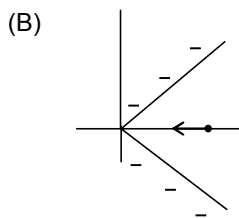
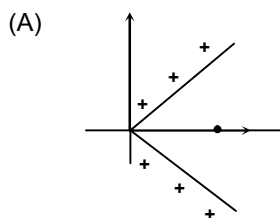
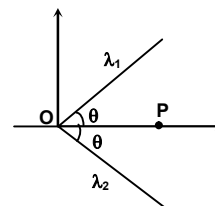
$$\vec{E}_R = (E_1 + E_2) \hat{i} - (E_1 + E_2) \hat{j}$$

$$= (E_1 + E_2) (\hat{i} - \hat{j})$$

$$\vec{E}_R = \frac{k\lambda}{a} [1 + \sqrt{2}] (\hat{i} - \hat{j})$$

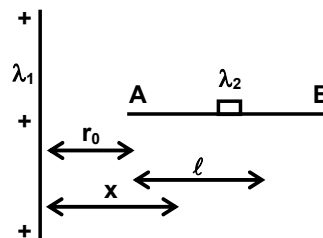


4. (A) → (p), (B) → (r), (C) → (s), (D) → (q)



$$5. \quad \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0} \ln \left(1 + \frac{\ell}{r_0} \right)$$

$$\begin{aligned} 5. \quad dF &= E (\lambda_2 dx) \\ &= \frac{2k\lambda_1}{x} \lambda_2 d\eta \\ &= 2k\lambda_1 \lambda_2 [\ell \ln \eta]_{r_0} \\ &= 2k\lambda_1 \lambda_2 [\ell \ln(r_0 + \ell) - \ell \ln r_0] \\ &= 2k\lambda_1 \lambda_2 \left[\ell \ln \left(1 + \frac{\ell}{r_0} \right) \right] \\ &= \frac{\lambda_1 \lambda_2}{2\pi\epsilon_0} \ell \ln \left(1 + \ell / r_0 \right) \end{aligned}$$



$$6. \quad 2$$

$$6. \quad E = \frac{k\ell\eta}{(R^2 + X^2)^{3/2}}$$

$$\text{Ra } x \gg R$$

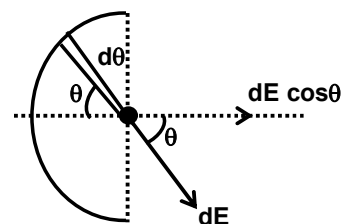
$$E = \frac{k\ell x}{x^3 x^2} = \frac{k\ell}{x^2}$$

7. (a) $\frac{2k\lambda}{R}$, (b) $E_{axis} = \frac{k\pi\lambda Rx}{(R^2 + x^2)^{3/2}}$; $E_{\perp} = \frac{2k\lambda R^2}{(R^2 + x^2)^{3/2}}$

7. $dE_x = dE \cos\theta$
 $= \frac{kd^2}{R^2} \cdot \cos\theta$
 $= \frac{k \cdot \lambda}{R} R / d\theta \cdot \cos\theta$

$$\int dE_x = \frac{k\lambda^2}{R} \int_{-\pi/2}^{\pi/2} \cos\theta d\theta$$

$$E_x = \frac{k\lambda}{R} \cdot [\sin\theta]_{-\pi/2}^{\pi/2} = \frac{k\lambda}{R} \cdot 2 = \frac{2k\lambda}{R}$$



θ

8. $F_{max} = \frac{Qq}{6\sqrt{3}\pi\epsilon_0 a^2}$

8. $E = \frac{kQ \cdot x}{(R^2 + x^2)^{3/2}}$

$$\frac{dE}{dx} = kQ \left[1 \cdot \frac{(R^2 + x^2)^{3/2} - \frac{3}{2}(R^2 + x^2)^{1/2} \cdot 2x^2}{(R^2 + x^2)^3} \right]$$

$$(R^2 + x^2)^{1/2} [R^2 + x^2 - 3x^2] = 0$$

$$x = \frac{R}{\sqrt{2}}$$

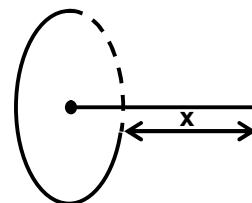
$$F = Eq$$

$$= \frac{k\theta \cdot R / \sqrt{2}}{\left(R^2 + \frac{R^2}{2}\right)^{3/2}} \cdot q$$

$$= \frac{k\theta q \cdot R}{\sqrt{2} \left(\frac{3R^2}{2}\right)^{3/2}}$$

$$= \frac{1}{4\pi\epsilon_0} \cdot \frac{\theta q \cdot R 2\sqrt{2}}{\sqrt{23}\sqrt{3} \cdot R^3 R^2}$$

$$F = \frac{\theta q}{6\pi\sqrt{3}\pi\epsilon_0}$$

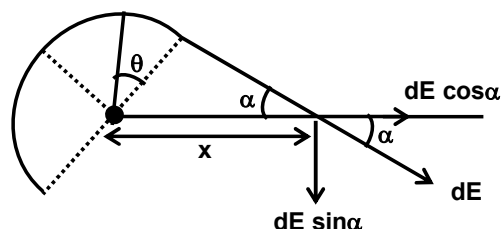


(b) $dE_x = dE \cos^2\theta$
 $= \frac{k(\lambda R d\theta)}{(x^2 + R^2)} \cdot \frac{x}{\sqrt{R^2 + x^2}}$
 $dE_x = \frac{k\lambda R_x}{(R^2 + x^2)^{3/2}} \int_{-\pi/2}^{\pi/2} d\theta$
 $= \frac{\lambda \cdot R \cdot x}{4\pi\epsilon_0 (R^2 + x^2)^{3/2}} \cdot \pi$

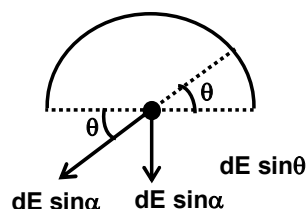
$$E_f = \frac{\lambda R \cdot}{4\epsilon_0 (R^2 + x^2)^{3/2}}$$

$$dE_y = dE \sin\alpha \sin\theta$$

$$= \frac{k(\lambda R d\theta)}{(R^2 + x^2)} \cdot \frac{R}{\sqrt{R^2 + x^2}} \cdot \sin\theta$$



(1)



$$\begin{aligned}
 &= \frac{k\lambda R^2}{(R^2 + x^2)^{3/2}} \cdot \sin\theta d\theta \\
 &= \frac{\lambda R^2}{4\pi k\epsilon_0 (R^2 + x^2)^{3/2}} \cdot \int_{-\pi/2}^{\pi/2} \sin\theta d\theta \\
 &= \frac{\lambda R^2}{4\pi \epsilon_0 (R^2 + x^2)^{3/2}} \cdot 2 \\
 E_y &= \frac{\lambda R^2}{2\pi \epsilon_0 (R^2 + x^2)^{3/2}} \quad (2)
 \end{aligned}$$

9. $E_{c1} = 0$; $E_{c2} = \frac{k\lambda}{R}$ (upward)

9. $E_2 = 2$

$$= 2 \cdot \frac{k\lambda}{R} [\cos\beta - \cos\alpha]$$

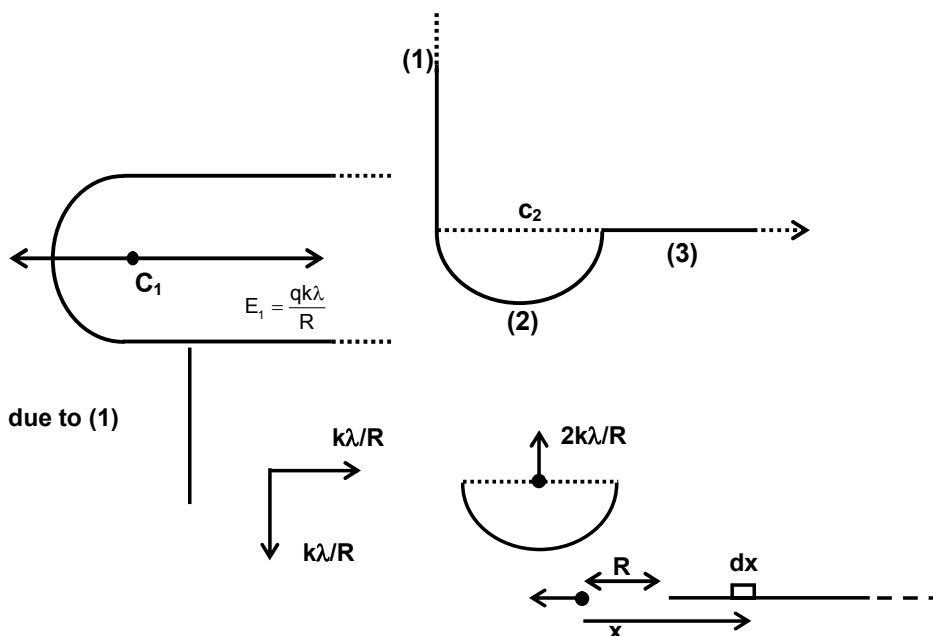
$$E_2 = 2 \cdot \frac{k\lambda}{R} \{1 - 0\}$$

$$E_{\text{net}} = \frac{2k\lambda}{R} - \frac{2k\lambda}{R} = 0$$

$$\int_2^{\infty} dE = \int \frac{k\lambda dx}{x^2}$$

$$E_2 = -k\lambda \left[\frac{1}{x} \right]_R^{\infty}$$

$$E_2 = \frac{k\lambda}{R}$$



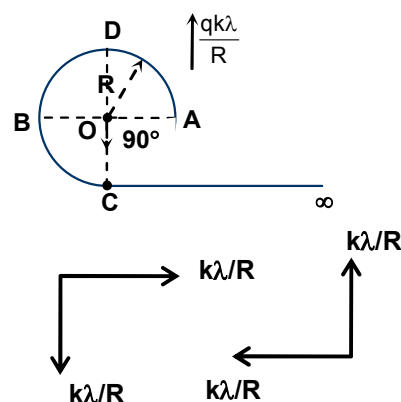
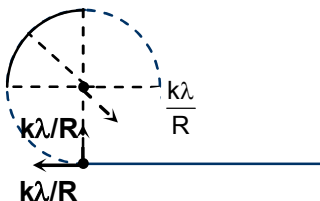
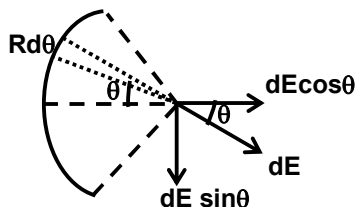
10. zero

10. Electric field due to AD & BC is cancelled

$$dE_x = dE \cos\theta$$

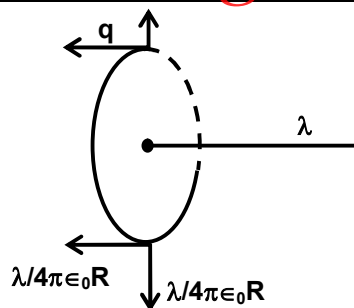
$$dE_x = \frac{k(\pi R d\theta)}{R^2} \cos\theta$$

$$E_x = \frac{k\lambda}{R} \cdot (\sin\theta)_{-\pi/4}^{\pi/4} = \frac{k\lambda}{R} \cdot \frac{2}{\sqrt{2}} = \frac{k\lambda}{R} \sqrt{2}$$



11. $\frac{q\lambda}{4\pi\epsilon_0 R}$

$$11. \quad F = \frac{\lambda}{4\pi\epsilon_0 R} \times q$$



$$12. \quad T = 2\pi \sqrt{\frac{2\epsilon_0 m a^2}{\lambda Q}}$$

$$12. \quad F = \frac{kq}{(a^2 + x^2)^{3/2}} \times Q \quad q = \lambda \cdot 2\pi a$$

$$Fm\alpha = \frac{1}{4\pi\epsilon_0} \cdot \frac{\lambda \cdot x \cdot \theta}{(a^2 + x^2)^{3/2}}$$

$$x \ll a$$

$$\alpha = \frac{\lambda a \theta}{q m a^3 \epsilon_0} x$$

$$\alpha = - \left(\frac{\lambda \theta}{\epsilon_0 q m a^2} \right) \cdot x$$

$$\alpha = - \cos^2 x$$

$$\frac{q\pi}{T} = \omega = \sqrt{\frac{\lambda Q}{q m a^2 \epsilon_0}}$$

$$T = q\pi \sqrt{\frac{q m a^2 \epsilon_0}{Q}}$$

$$13. \quad \tan^{-1} \left(\frac{\sigma q}{2\epsilon_0 m g} \right)$$

In equilibrium, along x-axis,

$$T \sin \theta = qE$$

$$\Rightarrow T \sin \theta = q \frac{\sigma}{2\epsilon_0} \quad \dots (1)$$

where T is the tension in the string.

Along y-axis in equilibrium,

$$T \cos \theta = mg \quad \dots (2)$$

$$14. \quad 1.2 \mu C/m^2$$

$$14. \quad \int dE = \int \frac{k(\sigma q \pi r dr) x}{(r^2 + \lambda^2)^{3/2}}$$

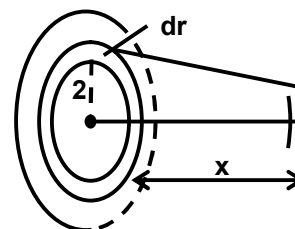
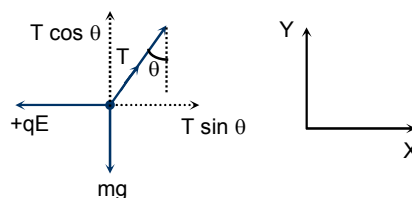
$$E = 2\pi k \sigma x \int_0^R \frac{r dr}{(r^2 + \lambda^2)^{3/2}}$$

$$r^2 + \lambda^2 = 4^2$$

$$kr dr = kudy$$

$$E = 2\pi k \sigma \cdot x \int \frac{4 dy}{4^3 4^2}$$

$$= q\pi k \sigma \cdot x \left(-\frac{1}{4} \right)_0^R$$



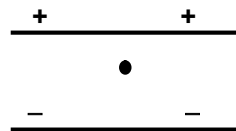
$$= -q\pi k\sigma \cdot x \left(\frac{1}{4} \right)_0^R$$

$$= -q\pi k\sigma \cdot x \left(\frac{1}{\sqrt{R^2 + x^2}} - \frac{1}{R} \right)$$

$$E = \frac{q\pi \times 1}{4\pi \epsilon_0} \cdot \sigma \cdot x \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + x^2}} \right]$$

15. Proportional to $(\sigma_A + \sigma_B)/2\epsilon_0$

15. $\frac{\sigma_A + \sigma_B}{2\epsilon_0}$



16. $A \rightarrow (q), B \rightarrow (p), C \rightarrow (s), D \rightarrow (p, r)$

17. D

LEVEL - II

1. $\frac{q}{\pi^2 \epsilon_0 r^2}$

1. $dE_1 = qdE \sin\theta$

$$= q \cdot \frac{k dq}{R^2} \cdot \sin\theta$$

$$= 2 \frac{k \cdot (\lambda R d\theta)}{R^2} \cdot \sin\theta$$

$$= -\frac{2k\lambda}{R} \cdot [\cos\theta]_0^{\pi/2} = \frac{2k\lambda}{R}$$

$$= \frac{2 \times 1}{4\pi \epsilon_0 R} \cdot \frac{q}{\pi R}$$

$$= \frac{q}{\pi^2 \epsilon_0 R^2}$$

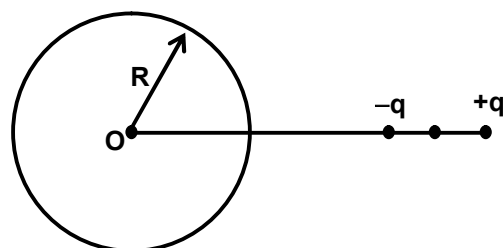
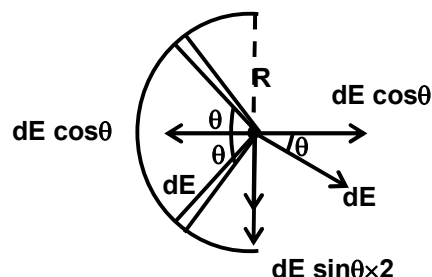
$$\lambda = \frac{q \times 2}{\pi}$$

2. $\frac{Qqa}{\pi \epsilon_0 x^3}$

2. $v = pE$

$$v = \frac{kQx}{(R^2 + x^2)^{3/2}}$$

$$v = \frac{1}{4\pi \epsilon_0} \cdot \frac{q \cdot qa q \cdot x}{(R^2 + x^2)^{3/2}}$$



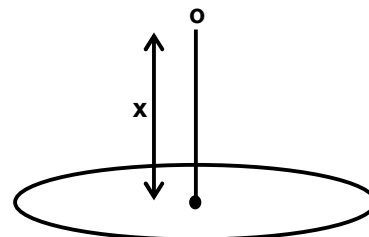
3. (A) $\rightarrow (q, r, s)$ (B) $\rightarrow q, r, s$ (C) $\rightarrow (r)$ (D) $\rightarrow (q, r)$

3. $\frac{\sigma x}{q \epsilon_0} \left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + x^2}} \right] = mg$

$$\frac{\sigma \times 0.1}{2f_0} \cdot \left[\frac{1}{0.1} - \frac{1}{\sqrt{10^{-2} + 10^{-2}}} \right] 100 \times 10^{-6} = 0.2 \times 10$$

$$\frac{\sigma \times 0.1}{q \epsilon_0} \cdot \left[\frac{1}{0.1} - \frac{1}{\sqrt{2 \times 0.1}} \right] \times 100 \times 10^{-6} = 2$$

$$\frac{\sigma \times 0.1}{2 \epsilon_0} \cdot \frac{1}{0.1} \cdot \left[10 \frac{1}{\sqrt{2}} \right] \times 10^{-4} = 2$$



$$\sigma = \frac{2 \times \sqrt{2} \times 10^4}{(\sqrt{2} - 1)} \times 2 \times 8.86 \times 10^{-12}$$

$$= \frac{2 \times 1.414 \times 8.86 \times 2 \times 10^{-8}}{(0.414)}$$

$$= 121.04 \times 10^{-8}$$

$$= 1.21 \mu\text{C} / \text{m}^2 = 1.2 \mu\text{C} / \text{m}_2$$

4. $\frac{Qr_1^2}{4\pi \epsilon_0 R^4}$

4. $e = \frac{Q}{\pi R^4} r$

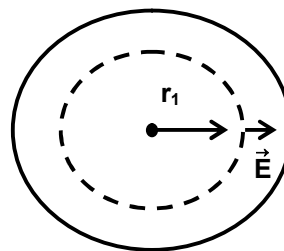
$$dq = e \cdot 4\pi r^2 \cdot dr$$

$$dq = \frac{Q}{\pi R^4} r \cdot 4\pi r^2 dr$$

$$q = \int dq = \int_0^r \frac{Q}{R^4} r^3 dr \Rightarrow q = \frac{4Q}{R^4} \cdot \frac{r^4}{4} = \frac{Qr^4}{R^4}$$

$$E 4\pi r^2 = \frac{Qr^4 r^2}{\epsilon_0 R^4}$$

$$E = \frac{Qr^2}{4\pi \epsilon_0 R^4}$$



5. R

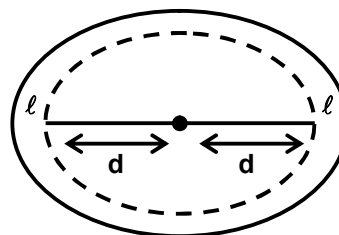
5. $\frac{k\ell^2}{4d^2} = \frac{k \cdot q' \cdot x}{R^3} \cdot \ell$

$$\frac{\ell^2}{4d^2} = \frac{q\ell d^3}{R^3} \cdot \frac{x}{R^3} \cdot \ell$$

$$\frac{1}{8} \times R^6 = d^6$$

$$d = \frac{R}{2}$$

$$qd = R$$



6. $\frac{k\lambda_0}{L} (\ln(2) - 1)$

6. $dE = \frac{k dq}{(x + 2L)^2}$

$$dE = \frac{k \cdot \lambda dx}{(x + 2L)^2}$$

$$dE = \frac{k \cdot \lambda_0 x}{\ell (x + 2L)^2}$$

kr

$$= \frac{k\lambda_0}{\ell} \cdot \frac{(4 - 2L)}{4^2} dx$$

x+2L=4

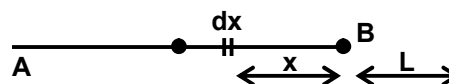
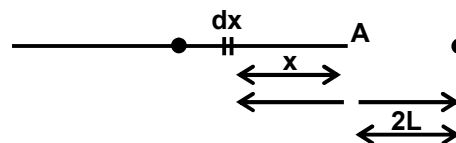
$$= \frac{k\lambda_0}{\ell} \left[\int_0^1 \frac{1}{4} d4 - 2L \int_0^1 \frac{d4}{4^2} \right]$$

dx = d4

$$= \frac{k\lambda_0}{\ell} \cdot \left[\ell n4 + 2L \frac{x^1}{u} \right]$$

$$dq = \lambda \cdot dx$$

$$dE = \frac{k dq}{(x + 2)^2} d$$



$$dE = \frac{k\lambda_0}{L} \int \frac{ndx}{(x+L)^2}$$

$$\text{let } 4 = x + L$$

$$d4 = dx$$

$$dE = \frac{k\lambda_0}{L} \int \frac{(4-L)d4}{4^2}$$

$$= \frac{k\lambda_0}{L} \cdot \left[\int \frac{1}{4} d4 - L \int \frac{1}{4^2} \right]$$

$$= \frac{k\lambda_0}{L} \cdot \left[\ln 4 + \frac{L}{4} \right]$$

$$= \frac{k\lambda_0}{\lambda} \cdot \left[\ln(x+L) \right]_0^L + L \left[\frac{1}{x+L} \right]_0^L$$

$$= \frac{k\lambda_0}{\lambda} \left[\ln 2L + L \cdot \left\{ \frac{1}{2L} - \frac{1}{L} \right\} \right]$$

$$= \frac{k\lambda_0}{\lambda} \left[\ln 2L - \frac{1}{2} \right]$$

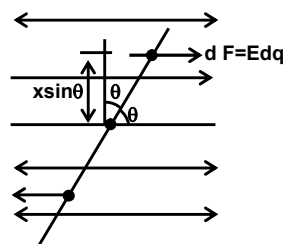
$$7. \frac{\sigma \lambda \ell^2}{2\epsilon_0} \sin \theta$$

$$7. dC = dF \cdot x \sin \theta \times 2$$

$$= \frac{\sigma}{q \epsilon_0} (\lambda \int dx) (\sin \theta \cdot x) \times 2$$

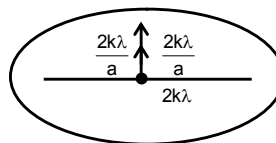
$$= q \times \frac{\lambda \sigma}{2 \epsilon_0} \cdot \sin \theta \cdot \left[\frac{x^2}{2} \right]_0^\ell$$

$$\tau = \frac{\lambda \sigma}{2 \epsilon_0} \cdot \ell^2 \sin \theta$$



$$8. E_{net} = \frac{4k\lambda}{a}$$

$$8. \frac{4k\lambda}{\sigma}$$



$$9. B, D$$

Time period = 4 T(SHM) and is independent of d.

$$\text{So if } OB = 2d \text{ coverage of AO requires } \frac{1}{12} \text{ th of time period} = \frac{4T}{12} = \frac{T}{3}$$

$$\text{and if } OB = \sqrt{2} d \text{ coverage of AO requires } \frac{1}{8} \text{ th of time period} = \frac{4T}{8} = \frac{T}{2}$$

Time period is inversely proportion to ω

$$\text{and } \omega^2 \propto \frac{Qq}{R^3} \left(\because \text{for small } d \frac{Qqd}{4\pi\epsilon_0(d^2 + R^2)^{3/2}} \approx \frac{Qqd}{4\pi\epsilon_0 R^3} \right)$$

$$\therefore \text{ if } Q \text{ doubles, } \omega \text{ becomes } \sqrt{2} \text{ times and } T \text{ becomes } \frac{1}{\sqrt{2}} \text{ times.}$$

also if R doubles, T becomes $2\sqrt{2}$ times.

$$10. \frac{-2\sigma}{\epsilon_0} \hat{k}$$

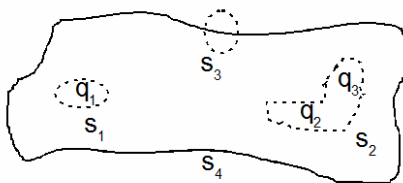
$$10. \frac{\sigma}{2\epsilon_0} + \frac{2\sigma}{2\epsilon_0} + \frac{\sigma}{2\epsilon_0} = \frac{2\sigma}{\epsilon_0}$$

CPP

ELECTROSTATICS -SHEET: 7(Lecture – 7)

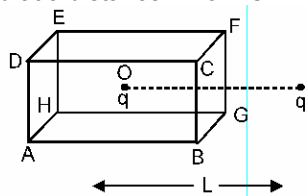
LEVEL – I

1. A rectangular surface of 2 metre width and 4 metre length, is placed in an electric field of intensity 20 newton/C, there is an angle of 60° between the perpendicular to surface and electrical field intensity. Find the total flux emitted from the surface. (in Volt- metre)
2. Mark the correct options:
 (A) Gauss's law is valid only for symmetrical charge distributions.
 (B) Gauss's law is valid only for charges placed in vacuum
 (C) The electric field calculated by Gauss's law is the field due to all the charges.
 (D) The flux of the electric field through a closed surface due to all the charges is equal to the flux due to the charges enclosed by the surface.
3. When no charge is confined with in the Gauss's surface, it may implies that-
 (A) $E = 0$
 (B) $\vec{E}$ and $d\vec{s}$ are parallel
 (C) $\vec{E}$ and $d\vec{s}$ are mutually perpendicular (D) $\vec{E}$ and $d\vec{s}$ are inclined at some angle
4. If the electric flux entering and leaving an enclosed surface respectively is ϕ_1 and ϕ_2 , find the electric charge inside the surface.
5. For which of the following fields, Gauss's law is valid-
 (A) fields following square inverse law (B) uniform field
 (C) all types of field (D) this law has no concern with the field
6. If electric field flux coming out of a closed surface is zero, the electric field at the surface will be-
 (A) zero (B) same at all places
 (C) dependent upon the location of points (D) infinites
7. If three electric di-poles are placed in some closed surface, find the electric flux emitting from the surface.
8. A charge q is inside a closed surface and charge $-q$ is outside. Find the outgoing electric flux is-
9. Three charges $q_1 = 1\mu\text{C}$, $q_2 = 2\mu\text{C}$ and $q_3 = -3\mu\text{C}$ and four surfaces S_1 , S_2 , S_3 and S_4 are shown. Find the flux emerging through surface S_2 in $\text{N-m}^2/\text{C}$.

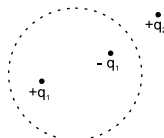


10. A square of side 20cm. is enclosed by a surface of sphere of 80 cm. radius. Square and sphere have the same centre. four charges $+2 \times 10^{-6} \text{ C}$, $-5 \times 10^{-6} \text{ C}$, $-3 \times 10^{-6} \text{ C}$, $+6 \times 10^{-6} \text{ C}$ are located at the four corners of a square, find the outgoing total flux from spherical surface in $\text{N-m}^2/\text{C}$.

11. A charged particle q is placed at the centre O of cube of length L (ABCDEFGH). Another same charge q is placed at a distance L from O . Find the electric flux through BCFG.



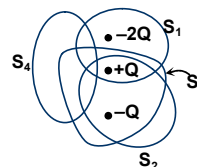
12. In the given figure, charges q_1 and $-q_1$ are inside a Gaussian surface. Where as charge q_2 is outside the surface. Electric field on the Gaussian surface will be



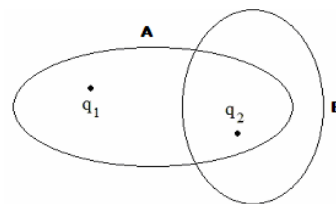
- (A) only due to q_2
 (B) zero on the Gaussian surface
 (C) uniform on the Gaussian surface
 (D) due to all

LEVEL – II

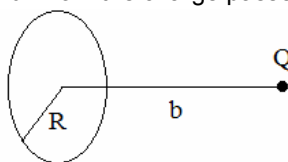
1. Four closed surfaces S_1 , S_2 , S_3 and S_4 together with the charges $-2Q$, Q and $-Q$ are shown in the figure. Find the electric flux through each surface.



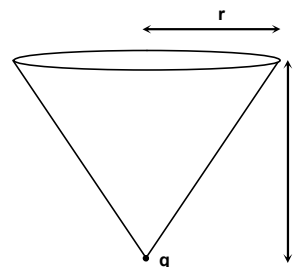
2. A charge of $1.2 \times 10^{-4} \text{ C}$ is placed inside a cylinder at the mid-point of the axis of cylinder. The flux through one end of the cylinder is $4.5 \times 10^6 \text{ N-m}^2/\text{C}$. What is the electric flux through the curved part of the cylindrical surface?
3. Charge is uniformly distributed in a space. The net flux passing through the surface of an imaginary cube of side "a" in the space is ϕ . What will be the net flux passing through the surface of an imaginary sphere of radius "a" in the space.
4. A charge Q is uniformly distributed over a rod of length l . Consider a hypothetical cube of edge l with the centre of the cube at one end of the rod. (a) Find the minimum possible flux of the electric field through the entire surface of the cube. (b) Find the maximum possible flux of the electric field through the entire surface of the cube.
5. Determine the electric flux through the closed surfaces A and B shown in the figure below. Take, $q_1 = 1.0 \times 10^{-10} \text{ C}$ and $q_2 = -1.0 \times 10^{-10} \text{ C}$.



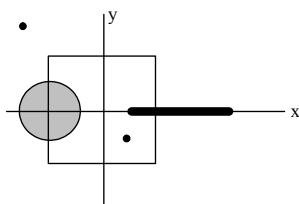
6. A point charge Q is located on the axis of a disc of radius R at a distance b from the plane of the disc (figure). Show that if one-fourth of the electric flux from the charge passes through the disc, then $R = \sqrt{3}b$.



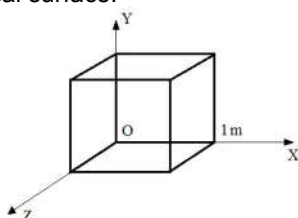
7. A point charge q is kept on the vertex of the cone of base radius r and height r . Find the electric flux through the curved surface.



8. A disk of radius $a/4$ having a uniformly distributed charge $6C$ is placed in the x - y plane with its centre at $(-a/2, 0, 0)$. A rod of length ' a ' carrying a uniformly distributed charge $8C$ is placed on the x -axis from $x = a/4$ to $x = 5a/4$. Two point charges $-7C$ and $3C$ are placed at $(a/4, -a/4, 0)$ and $(-3a/4, 3a/4, 0)$ respectively. Consider a cubical surface formed by six surfaces $x = \pm a/2, y = \pm a/2, z = \pm a/2$. Find the electric flux through this cubical surface.



9. The electric field everywhere on the surface of a thin spherical shell of radius 0.750 m is measured to be equal to 890 N/C and points radially towards the centre of the sphere. (a) What is the net charge within the sphere's surface? (b) Find the total electric flux passing through the shell.
10. A charge q is placed just above the centre of a horizontal circle of radius ' r ', and a hemisphere of this radius is erected about the charge. Compute the electric flux through the closed surface that consists of hemisphere and the plane circle, using Gauss's law. Verify this result without using Gauss's law.
11. Consider a cube of side ' $a = 1$ m' located at the origin. An electric field given by $\vec{E} = bx^2 \text{ N/C} \hat{i}$, where $b = 2.4 \times 10^4 \text{ V/m}^3$ is a constant, is present in space. Calculate the net electric flux through the cube. What is the net charge enclosed by the cubical surface.



12. The electric field in a particular space is $\vec{E} = (x+2) \text{ N/C} \hat{i}$ with x measured in meters. Consider a cylinder of radius = 20 cm that is coaxial with the x -axis. One end of the cylinder is positioned at $x = 0$ and the other end at $x = 2.0$ m. (a) what is the magnitude of the electric flux through the end of the cylinder at $x = 2.0$ m. (b) What net charge is enclosed within the cylinder?

CPP

ELECTROSTATICS -SHEET: 7(Lecture – 7)

ANSWER KEY

Level-I

1. 80
1. Area of rectangular surface $|\vec{A}| = 8 \text{ m}^2$
 $|\vec{E}| = 20 \text{ N/C} \quad \theta = 60^\circ$
 $\phi = \vec{E} \cdot \Delta \vec{A}$
 $= 8 \text{ m}^2 \cdot 20 \text{ N/C} \cdot \cos 60^\circ$
 $\phi = 80 \text{ v} - \text{m}$

2. C, D 3. A, C, D

4. $(\phi_2 - \phi_1)/\epsilon_0(\phi_2 - \phi_1) \epsilon_0$

4.
$$\phi_{\text{net}} = \phi_2 - \phi_1 = \frac{q_{\text{in}}}{\epsilon_0}$$

$$q_{\text{in}} = \epsilon_0 (\phi_2 - \phi_1)$$

5. A 6. C

7. Zero

7. Total charge enclosed by the surface = zero so, the electric flux net emitting through the surface will be zero.

8. q/ϵ_0 9. $-36\pi \times 10^3$

9.
$$\phi \text{ (through } S_2) = \frac{q_2 + q_3}{\epsilon_0} = \frac{2\mu\text{C} - 3\mu\text{C}}{\epsilon_0}$$

$$= -\frac{1\mu\text{C}}{4\pi\epsilon_0} \times 4\pi = -4\pi \times 10^{-6} \times k$$

$$\phi = -4\pi \times 10^{-6} \times 9 \times 10^9 = -36\pi \times 10^3 \text{ v} \cdot \text{m}$$

10. Zero

11. Zero

11. Total flux coming out through the cube by the enclosed charge = $\frac{q}{\epsilon_0}$ So, the total flux through BCFG = $\frac{q}{6\epsilon_0}$ (due to the charge q at o) and due to the charge q at P outside thecube through BCFG = $-\frac{q}{6\epsilon_0}$ so the net flux = $\frac{q}{6\epsilon_0} - \frac{q}{6\epsilon_0} = 0$

12. D

Level-II

1. $\phi_1 = \frac{-Q}{\epsilon_0}, \phi_2 = 0, \phi_3 = 0 \text{ and } \phi_4 = 0$

1.
$$\phi_1(S_1) = \frac{q_{\text{in}}}{\epsilon_0} = \frac{\theta}{\epsilon_0}$$

$$\phi_2(S_2) = \frac{q_{\text{in}}}{\epsilon_0} = \text{zero}$$

$$\phi_3(S_3) = \frac{q_{\text{in}}}{\epsilon_0} = \text{zero}$$

$$\phi_4(S_4) = \frac{q_{\text{in}}}{\epsilon_0} = \text{zero}$$

2. $4.6 \times 10^6 \text{ N} \cdot \text{m}^2 / \text{C}$

2. Flux through one end of cylinder = $4.5 \times 10^6 \text{ nm}^2/\text{c}$ through other end will be also equal due to symmetry

$$\text{Total flux} = \frac{q_{\text{in}}}{\epsilon_0} = \frac{1.2 \times 10^{-4}}{4\pi\epsilon_0} \times 4\pi = 9 \times 10^9 \times 1.2 \times 10^{-4} \times 4\pi = 4\pi \times 10.8 \times 10^5$$

$$\text{Flux through the curved surface} = \phi_{\text{net}} - \text{total flux through the end} \Rightarrow 4\pi \times 10.8 \times 10^5 - 9 \times 10^6$$
$$13.6 \times 10^6 - 9 \times 10^6 = 4.6 \times 10^6 \text{ Nm}^2 / \text{C}$$

3. $\frac{4\pi}{3} \phi$

3.
$$\phi_0 = \frac{q_{\text{in}}}{\epsilon_0}$$

$$\phi_0 = \frac{\rho \times \frac{4}{3} \pi R^3}{\epsilon_0}$$

$$\phi_0 = \frac{\rho \times \frac{4}{3} \pi a^3}{\epsilon_0}$$

$$\phi_0 = \frac{4}{3} \pi \times \frac{\phi \epsilon_0}{\epsilon_0}$$

$$\phi_0 = \frac{4\pi}{3} \phi$$

Flux through the cube

$$\phi = \frac{q_{in}}{\epsilon_0}$$

$$\phi = \frac{\rho a^3}{\epsilon_0}$$

$$\rho a^3 = \phi \epsilon_0$$

4. (a) $Q/2\epsilon_0$ (b) $Q/\sqrt{2}\epsilon_0$
 4. Flux will be maximum when the rod will be placed along the diagonal of cube. Total length of the rod along the diagonal keeping one end at centre inside the cube = $\frac{\ell}{\sqrt{2}}$

(a) ϕ_{min} (when the rod placed || to the edge) = $\frac{\lambda \frac{\ell}{2}}{\epsilon_0} = \frac{\phi}{\ell} \times \frac{\ell}{2} \times \frac{1}{\epsilon_0}$

$$\phi_{min} = \frac{\theta}{2 \epsilon_0}$$

(b) $\phi_{max} = \frac{q_{in}}{\epsilon_0} = \frac{\lambda \frac{\ell}{\sqrt{2}}}{\epsilon_0} = \frac{\theta}{\ell} \times \frac{\ell}{\sqrt{2}} \times \frac{1}{\epsilon_0}$ $\phi_{max} = \frac{\theta}{\sqrt{2} \epsilon_0}$

5. $\phi_A = 0$; $\phi_B = 11.3V - m$

5. $\phi_A = \frac{q_{in}}{\epsilon_0} = \frac{q_1 + q_2}{\epsilon_0} = 0$

$$\phi_B = \frac{q_{in}}{\epsilon_0} = \frac{q_2}{\epsilon_0} = \frac{4\pi q_2}{4\pi \epsilon_0} = 9 \times 10^9 \times 4\pi \times (-10^{10}) = 11.3V - m.$$

6. Let assume $R = \sqrt{3}b$ $\tan \alpha = \frac{\sqrt{3}b}{b} = \sqrt{3}$ $d = 60^\circ$

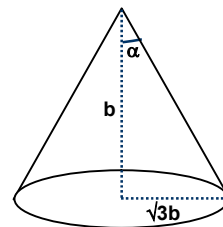
Solid angle $\Omega = 2\pi (1 - \cos \alpha)$

$$\Omega = 2\pi (1 - \cos 60^\circ)$$

$$= 2\pi \left(1 - \frac{1}{2}\right) = \pi$$

Total flux due to Pt. charge $\theta = \frac{\theta}{\epsilon_0}$ when the solid angle is 4π so, the flux for π solid

$$\text{angle} = \frac{\theta}{4 \epsilon_0} \text{ proved.}$$



7. zero

8. $\frac{-2C}{\epsilon_0}$

$$8. \quad \phi = \frac{q_{in}}{\epsilon_0} = \frac{q_{rod} + q_{disc} + q_{Pt-c}}{\epsilon_0}$$

$$\phi = \frac{2c + 3c + (-7c)}{\epsilon_0}$$

$$\phi = \frac{2c + 3c + (-7c)}{\epsilon_0} = \frac{-2}{\epsilon_0} c.$$

$$9. \quad (a) 5.6 \times 10^{-8} C \quad (b) 6.3 \times 10^3 v - c$$

$$9. \quad |\vec{E}| = 890 N/C$$

$$r = 0.75 m$$

(a) angle b/w $\vec{E}$ and area vector is π

$$\phi = -E \Delta S = \frac{q_{in}}{\epsilon_0}$$

$$Q_{in} = -E (4\pi \epsilon_0) r^2$$

$$= -890 \times \frac{9}{16} \times \frac{1}{9 \times 10^9}$$

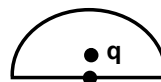
$$= -5.6 \times 10^{-8} C$$

$$(b) \quad \phi = \frac{q_{in}}{\epsilon_0} = -\frac{5.6 \times 10^{-8} \times 4\pi}{4\pi \epsilon_0}$$

$$\phi = -6.3 \times 10^3 v - c$$

$$10. \quad q/\epsilon_0$$

$$10. \quad \Phi = \frac{q_{in}}{\epsilon_0} = \frac{q}{\epsilon_0}$$



$$11. \quad \phi = 2.4 \times 10^4 v - m \quad q_{in} = \epsilon_0 \phi$$

$$11. \quad \phi_{net} \cdot \vec{E} \cdot \Delta \vec{S}$$

$$\Delta \vec{S} \text{ through the P/ane } (x = 1) = 1 m^2$$

$$\vec{E} = b \hat{i} = 0$$

$$\phi_{net} = b v - m$$

$$= 2.4 \times 10^4 v - m$$

$$q_{net} = \phi \epsilon_0$$

$$12. \quad (a) \quad \phi = \vec{E} \cdot \Delta \vec{S}$$

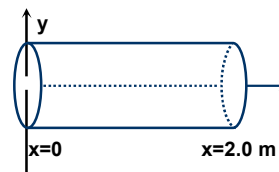
$$= (2+2) \hat{i} \cdot \pi (0.2)^2 \hat{i} \frac{Nm^2}{C} = 0.16\pi \frac{Nm^2}{C}$$

$$(b) \quad \phi = \vec{E} \cdot d\vec{S} = \frac{q_{in}}{\epsilon_0}$$

$$\therefore \phi_{x=2} + \phi_{x=0} = \frac{q_{in}}{\epsilon_0}$$

$$\therefore 0.16\pi - 2(\pi)(0.2)^2 = \frac{q_{in}}{\epsilon_0}$$

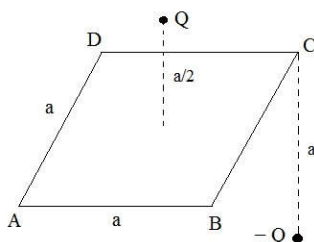
$$\therefore q_{in} = (0.08\pi \epsilon_0) C$$



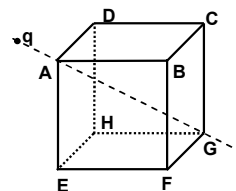
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ELECTROSTATICS -SHEET: 8(Lecture – 8)

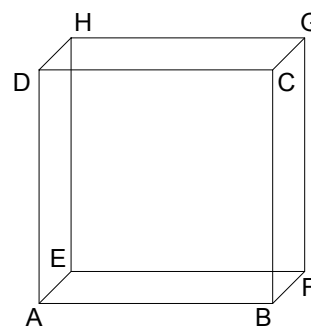
1. A given region has an electric field that is the vector sum of two different fields, a field due to a charge $q = 5 \times 10^{-8} \text{ C}$ at the origin plus a uniform field of strength $E_0 = 3000 \text{ N/C}$ in the x-direction. Calculate the flux through each side of a cube with side of length 20 cm each and parallel to x-, y- and z- directions, the cube is centered at the origin.
2. Consider two point charges Q and $-Q$. Find the total electric flux through the square plate shown in the figure.



3. A point charge q is located at a distance d from an infinite plane. Determine the electric flux through the plane due to point charge.
4. The line AG in the shown figure is a diagonal of a cube. A point charge q is located on the extension of line AG very close to vertex A of the cube. The electric flux passing through side ABCD is $-q/6k\epsilon_0$. Find the value of k .



5. A charge Q is kept on the Diagonal EC of the cube kept very close to E. The flux with outer surface EFBA of the cube is $\frac{nQ}{24\epsilon_0}$. Find value of n .



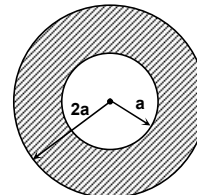
6. A uniformly charged and infinitely long line having a linear charge density ' λ ' is placed at a normal distance y from a point O. Consider a imaginary sphere of radius R with O as centre and $R > y$. Calculate the electric flux through the surface of the sphere.



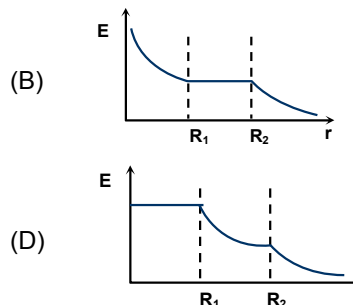
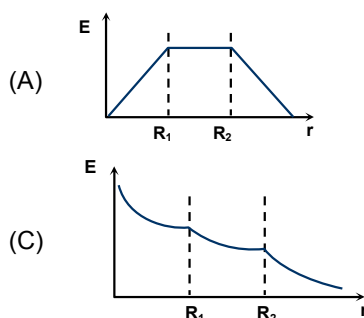
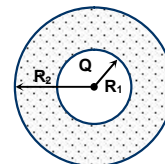
7. The electric flux due to a charge Q placed at the mouth of a conical flask is ϕ :
 (A) $\phi = 0$ (B) $\phi > \frac{Q}{2\epsilon_0}$ (C) $\phi > \frac{Q}{\epsilon_0}$ (D) $\phi = \frac{Q}{2\epsilon_0}$
8. A non-conducting uniformly charged sphere having volume charge density ' ρ ' have a spherical cavity. The centre of the cavity is displaced with respect to the centre of the ball by a distance ' a '. Find the electric field strength inside the cavity. Show that it is independent of the radius of cavity or the sphere.
9. Inside an infinitely long circular cylinder charged uniformly with volume density ρ there is a circular cylindrical cavity. The distance between the axis of cylinder and the cavity is ' a '. Find the electric field strength inside the cavity at a distance ' r ' from the axis of cylinder. The permittivity is assumed to be unity.

10. A spherical conductor has a radius of R and charge Q . A spherical shell of thickness R and uniform charge Q , is kept so as to be concentric to the conductor and touching the conductor. Find the electric field at a distance of $1.5 R$ from the center of the sphere.
11. Near the surface of the earth, there is a downward directed electric field of 150 Vm^{-1} . Assume the earth is a conducting medium and the atmosphere around earth is charged with uniform charged density ρ . At 200 m above the surface of the earth, the downward field is 100 Vm^{-1} . The value of ρ is $\epsilon_0/k \text{ C/m}^3$, where ϵ_0 denotes the electric permittivity of free space. Find the value of $|k|$.

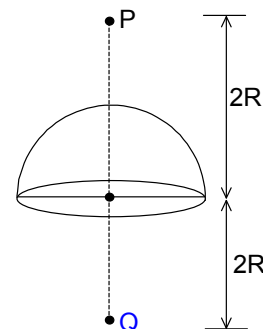
12. Consider a non conducting sphere of radius a carrying uniformly distributed charge q surrounded by a spherical shell of radius $2a$. The region between $a \leq r \leq 2a$ is filled with charge density $\rho = C/r$, where r is the distance from the centre. If the magnitude of electric field in the region $a \leq r \leq 2a$ is constant, then the magnitude of charge in the volume $a \leq r \leq 2a$ can be written as nq . Find the value of n ?



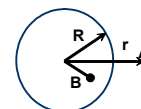
13. In a spherical distribution the charge density varies as $\rho(r) = \frac{Q}{2\pi R_1^2 r}$ for $R_1 < r < R_2$ as shown in the figure. A point charge Q lies at the centre of the sphere at $r = 0$. Choose the appropriate curve for variation of electric field E in the space as a function of r



14. Charge is distributed uniformly throughout the volume of an infinitely long cylinder of radius R . Find the electric field at a distance r from the cylinder axis (for $r < R$).
15. If the electric field due to uniform thin hemispherical shell at point P is E , find the magnitude of electric field at Q . (charge on the hemisphere is q_0 and radius R)



16. Figure shows a spherical symmetric distribution of charge (q) of radius R . Find electric field $\vec{E}$ for points A and B which are lying outside and inside the charge distribution, respectively.



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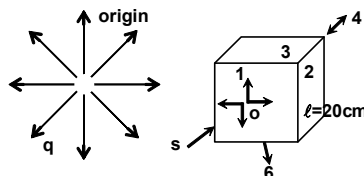
ELECTROSTATICS -SHEET: 8(Lecture – 8)

ANSWER KEY

$$1. \quad \phi_1 = \phi_4 = \phi_3 = \phi_6 = \frac{2}{6\epsilon_0} \xrightarrow{E_0 = 3000 \frac{N}{C}}$$

$$\phi_2 = \frac{2}{6\epsilon_0} + E_0 \cdot A$$

$$\phi_5 = \frac{2}{6\epsilon_0} - E_0 \cdot A$$



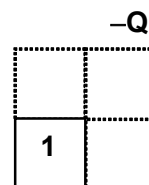
$$2. \quad 5Q/24\epsilon_0$$

$$2. \quad \phi_Q = \frac{Q}{6\epsilon_0}$$

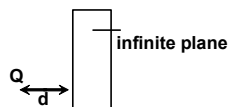
$$\phi_{-Q} \text{ through one plate surface } \frac{Q}{6\epsilon_0}$$

$$\phi_{-Q} \text{ through 1 only } = \frac{Q}{24\epsilon_0}$$

$$\text{Total flux through tan} = \frac{Q}{6\epsilon_0} + \frac{Q}{24\epsilon_0} = \frac{5Q}{24\epsilon_0}$$



$$3. \quad \text{Flux through the plane} = \frac{Q}{2\epsilon_0}$$



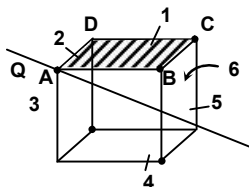
$$4. \quad 4$$

$$\phi \text{ through cube} = 0$$

$$\phi_4 = \phi_5 = \phi_6 = \frac{\theta}{24\epsilon_0}$$

$$4\phi_1 = \phi_2 = \phi_3 \quad \epsilon\phi_1 = 0$$

$$= 2\phi_1 = -3\phi_4 \Rightarrow \phi_1 = -\phi_4 = -\phi/24\epsilon_0 \Rightarrow k = 4$$



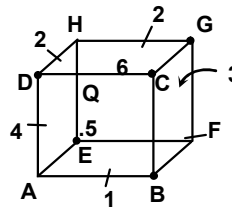
$$5. \quad 7$$

$$\epsilon\phi = \phi/\epsilon_0$$

$$\phi_2 = \phi_3 = \phi_5 = \frac{\theta}{24\epsilon_0}$$

$$\phi_1 = \phi_4 = \phi_6$$

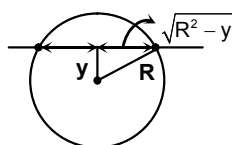
$$\Rightarrow 3\phi_1 + \frac{3Q}{4\epsilon_0} = \frac{Q}{\epsilon_0}$$



$$6. \quad \frac{2\lambda\sqrt{R^2 - y^2}}{\epsilon_0}$$

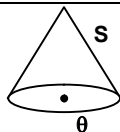
$$6. \quad \theta_{\text{equation}} = \lambda 2\sqrt{R^2 - y^2}$$

$$\Rightarrow \phi_s = \frac{2\lambda\sqrt{R^2 - y^2}}{\epsilon_0}$$



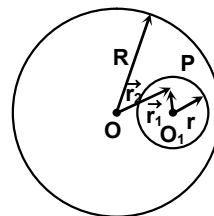
$$7. \quad Q/2\epsilon_0$$

7. $\phi_3 = \frac{Q}{2\epsilon_0}$



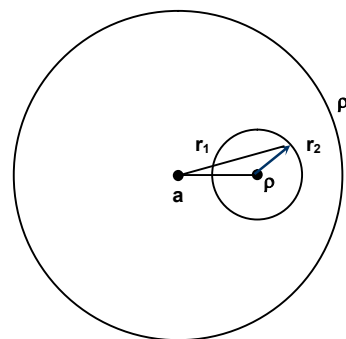
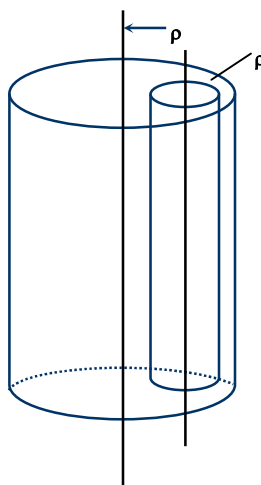
8. $\frac{a\rho}{3\epsilon_0}$

8. $\vec{E}_P = \vec{E}_{CS} + \vec{E}_{\text{cavity 1-vc charge}}$
 $= \frac{q\vec{r}_1}{3\epsilon_0} + \frac{(-q)\vec{r}_2}{3\epsilon_0}$
 $= \frac{q}{3\epsilon_0}(\vec{r}_1 - \vec{r}_2) = \frac{q}{3\epsilon_0}\vec{OO}$

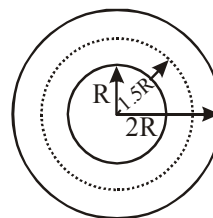


9. $\frac{a\rho}{2\epsilon_0}$

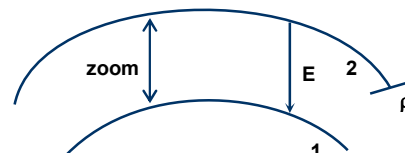
9. $\vec{E}_P = \vec{E}_P^{\text{comp.cyl}} \rightarrow \vec{E}_P^{\text{(Cavity-ve charge)}}$
 $= \frac{\rho\vec{r}_1}{2\epsilon_0} + (-\rho)\frac{\vec{r}_2}{2\epsilon_0}$
 $= \frac{\rho(\vec{r}_1 - \vec{r}_2)}{2\epsilon_0} = \frac{\rho\vec{a}}{2\epsilon_0}$



10. Total flux through this area is
 $= E \cdot 4\pi (1.5R)^2$
 $\frac{q_{\text{net}}}{\epsilon_0} = \frac{1}{\epsilon_0} \left(Q + \frac{Q \{ (1.5R)^2 - R^3 \}}{\{ (2R)^3 - R^3 \}} \right) = \frac{Q}{\epsilon_0} \left(1 + \frac{19}{56} \right) = \frac{75Q}{56\epsilon_0}$
 Gauss law $E \cdot 4\pi (1.5R)^2 = Q (75/56\epsilon_0)$
 $E = \frac{75Q}{504\pi\epsilon_0 R^2}$



11. 4
 $\frac{\rho(4\pi R^2 h)}{\epsilon_0} = 150 \left[(R+h)^2 - (R^2) \right] \times 100$
 $\frac{\rho R^2 \lambda}{\epsilon_0} = 1500(R+h)^2 - 100R^2$
 $\frac{\rho \lambda}{\epsilon_0} = 150 \left(1 + \frac{h}{R} \right)^2 - 100$
 $\frac{\rho h}{\epsilon_0} = 150 \left(1 + \frac{2h}{R} \right) - 100$
 $\frac{\rho h}{\epsilon_0} = 50 + \frac{300h}{R}$



$$\frac{\rho}{\epsilon_0} = \frac{500}{200} = \frac{1}{4} = \rho = \frac{\epsilon_0}{4}$$

$$k = 4$$

12. 3

$$q' = \int_a^r \frac{C}{r} (4\pi r^2) dr = 2\pi C(r^2 - a^2)$$

$$E = \frac{q + 2\pi C(r^2 - a^2)}{4\pi\epsilon_0 r^2} = \text{constant} \Rightarrow C = \frac{q}{2\pi a^2}$$

$$\therefore q' = 2\pi \left(\frac{q}{2\pi a^2} \right) [(2a)^2 - a^2] = 3q$$

13. B

13. For $r < R_1$

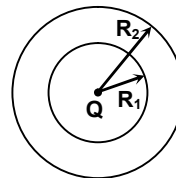
$$E = \frac{kQ}{r^2} \frac{Ax}{Dx}$$

For $R_2 > r > R_1$

$$E \cdot 4\pi r^2 = Q + \frac{\int_{R_1}^r \frac{Q}{2\pi R^2} \cdot 4\pi r dr}{\epsilon_0} = Q + \frac{\int_{R_1}^r 2Q \frac{rdr}{R^2}}{\epsilon_0}$$

$$= \frac{Q + Q \left(\frac{r^2 - R_1^2}{R^2} \right)}{\epsilon_0} = \frac{Qr^2}{\epsilon_0}$$

$$E = \frac{Q}{4\pi\epsilon_0} = emf$$



14. $\frac{\rho r}{2\epsilon_0}$

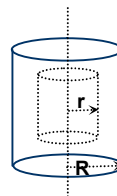
Consider a cylindrical element of radius 'r' and length 'ℓ'.

According to Gauss's Law

$$\int \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$

$$E(2\pi r\ell) = \frac{\rho(\pi r^2\ell)}{\epsilon_0}$$

$$E = \frac{\rho r}{2\epsilon_0}$$



15. (a) $\vec{E} = \frac{2k\lambda}{L} \left\{ \left(1 - \frac{1}{\sqrt{5}} \right) \hat{i} + \frac{2}{\sqrt{5}} \hat{j} \right\}$

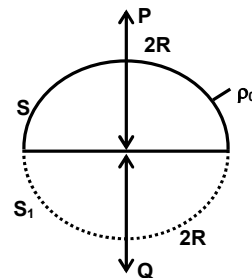
(b) $E = \frac{k\lambda}{2L}$

15. $E_s^p = E$

$E_{s_1}^p = E_s^q$

$E_{s_1}^p + E_s^p = K \frac{2\theta_0}{4R^2}$

$\Rightarrow E_{s_1}^p = \frac{K\rho_0}{2R^2} - E = E_s^q$



16. The spherically symmetric distribution of charge means that the charge density at any point depends only on the distance of the point from the centre and not on the direction. Secondly, the object cannot be a conductor, or else the excess charge will reside on its surface.

Now, applying Gauss' law to a spherical Gaussian surface of radius r ($r > R$ for point A),

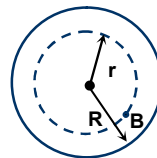
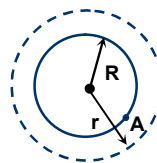
$$\epsilon_0 \oint \vec{E} \cdot d\vec{s} = q_{\text{in}} \Rightarrow \epsilon_0 E(4\pi r^2) = q$$

$$E = \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} \quad (\text{where } q \text{ is the charge enclosed in sphere of radius } r.)$$

For point B ($r < R$), $\epsilon_0 \oint \vec{E} \cdot d\vec{s} = \epsilon_0 E(4\pi r^2) = q'$

$$q' = \frac{q \frac{4}{3}\pi r^3}{\frac{4}{3}\pi R^3} = q \left(\frac{r}{R} \right)^3$$

$$\therefore E = \frac{1}{4\pi \epsilon_0} \frac{q}{R^3} \frac{(r/R)^3}{r^2} = \frac{1}{4\pi \epsilon_0} \frac{qr}{R^3}$$

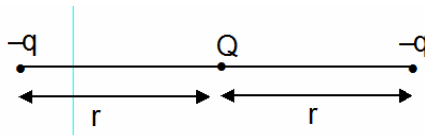


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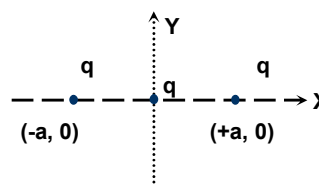
ELECTROSTATICS -SHEET: 9(Lecture – 9)

LEVEL – I

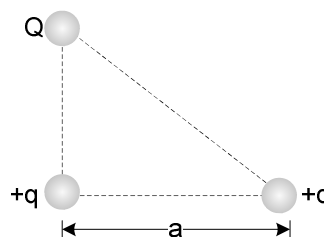
1. A point positive charge of Q' units is moved round another fixed point positive charge of Q units in circular path. If the radius of the circle r , find the work done on the charge Q in making one complete revolution.
2. A proton is projected with velocity 7.45×10^5 m/s towards another proton which is originally at rest. Find the minimum distance of approach.
3. Three charges are placed as shown in fig if the electric potential energy of system is zero, then find $Q : q$.



5. Two point charges each of charge $+q$ are fixed at $(+a, 0)$ and $(-a, 0)$. Another positive point charge q placed at the origin is free to move along x -axis. The charge q at the origin in equilibrium will have
 (A) maximum force and minimum potential energy
 (B) minimum force and maximum potential energy
 (C) maximum force and maximum potential energy
 (D) minimum force and minimum potential energy.



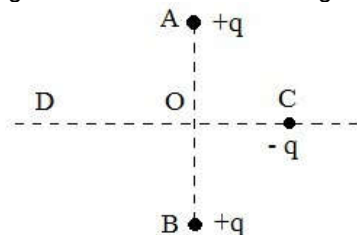
6. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right-angled isosceles triangle as shown. If the net electrostatic energy of the configuration is zero, calculate Q



7. Three point charge $-q$, $+q$ and $-q$ are placed along a straight line at equal distances(say r meter). If $+q$ charge is in the middle, find the Electric potential energy of this system of charges.
8. Four equal charges of charge q are placed at corner of a square of side a . Evaluate the Potential energy of the whole system.
9. Eight point charges with charge $+q$ on each of them is brought together to form a system with each charge placed at the vertices of a cube of each side ' a '. Find the minimum work required to do so in assembling the charges from infinity to form the system.
10. A charged particle ' q ' is shot towards another charged particle ' Q ', which is fixed, with a speed ' v '. It approaches ' Q ' upto a closest distance r and then returns. If q were given a speed of ' $2v$ ', find the closest distance of approach.
11. Two point charges $+q$ and $-q$ are held fixed at $(-d, 0)$ and $(d, 0)$ respectively of a (x, y) coordinate system , then -
 (A) The electric field $\vec{E}$ at all points on the x -axis has the same direction
 (B) $\vec{E}$ at all points on the Y - axis is along $\vec{i}$
 (C) Work has to be done in bringing a test charge from infinity to the origin
 (D) The dipole moment is $2qd$ directed along $\vec{i}$

Level-II

1. A point charge Q is placed at the centre of uniformly volume charged solid sphere of radius R having total charge Q . Find the total electrostatic energy of this system (ignore self energy of the point charge).
2. Positive and negative point charges of equal magnitude are kept at $\left(0, 0, \frac{a}{2}\right)$ and $\left(0, 0, -\frac{a}{2}\right)$, respectively. The work done by the electric field when another positive point charge is moved from $(-a, 0, 0)$ to $(0, a, 0)$ is
3. A hollow sphere of radius R is given uniformly distributed charge $+q$. If the sphere expand uniformly to a radius $2R$ and during expansion $q/2$ of its charge is extracted and removed from it. Calculate the work done by electric force in the system.
4. 2 identical charges $5 \mu\text{C}$ each are fixed at a distance 8 cm between them & a charged particle of mass $9 \times 10^{-6} \text{ kg}$ and $-10 \mu\text{C}$ is placed at a distance 5 cm from each of them & released. Calculate the speed of the particle when it is nearest to the two charges.
5. Four charges $+q, -q, +q$ and $-q$ are placed in order on the four consecutive corners of a square of side a . Find the work done in interchanging the positions of any two neighbouring charges of opposite sign.
6. Two fixed, equal and positive charges, each of magnitude $5 \times 10^{-5} \text{ C}$ are located at point A and B separated by a distance of 6 m . An equal and opposite charge moves towards them along the line COD, the perpendicular bisector of the line AB. The moving charge, when it reaches the point C at a distance of 4 m from O, has a kinetic energy of 4 Joules . Calculate the distance of the farthest point D which the negative charge will reach before returning towards C.



7. Two equal point charges are fixed at $x = -a$ and $x = +a$ on the x -axis. Another point charge Q is placed at the origin. The change in the electrical potential energy of Q , when it is displaced by a small distance x along the x -axis, is approximately proportional to x^a . Find a .

CPP**ELECTROSTATICS -SHEET: 9(Lecture – 9)****ANSWER KEY****Level-I**

1. Zero
1. $w = 0$ $F \perp$ disp.
2. 10^{-12} m
2. $U_i + K_i = U_f + K_f$

$$(p) \rightarrow v_0 \quad (p)$$

$$\frac{1}{2}mv_0^2 = 2 \times \frac{1}{2}m\left(\frac{v_0}{2}\right)^2 + \frac{k\ell^2}{r} \quad (p) \frac{\vec{v}_0}{2} \quad (p) \frac{\vec{v}_0}{2}$$

$$= \frac{ke^2}{r} = \frac{1}{4}mv_0^2$$

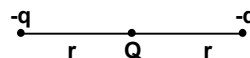
$$r = \frac{4ke^2}{mv_0^2} = \frac{4 \times 9 \times 10^9 \times (1.6 \times 10^{-19})^2}{1.67 \times 10^{-27} (7.45 \times 10^5)^2} = \frac{36 \times 1.6 \times 1.6 \times 10^{-29}}{1.6 \times 10^{-27} \times (7.45 \times 10^5)^2} = 10^{-12} \text{ m}$$

3. $\frac{Q}{q} = \frac{1}{4}$

$$U = \frac{k(-q)(-q)}{2r} + \frac{k(-q)\theta}{r} + \frac{k(-q)(\theta)}{r}$$

$$= \frac{kq}{2r} = \frac{2k\theta q}{r}$$

$$\frac{Q}{q} = \frac{1}{4}$$



5. (D)

The net force on q at origin is

$$\vec{F} = \vec{F}_1 + \vec{F}_2 = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{r^2} \hat{i} + \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{r^2} (-\hat{i}) = \hat{0}$$

The P.E. of the charge q in-between the extreme charges at a distance x from the origin along +ve x-axis is

$$U = \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{(a-x)} + \frac{1}{4\pi\epsilon_0} \cdot \frac{q^2}{(a+x)} = \frac{1}{4\pi\epsilon_0} \cdot q^2 \left[\frac{1}{a-x} + \frac{1}{a+x} \right]$$

$$\frac{dU}{dx} = \frac{q^2}{4\pi\epsilon_0} \left[-\frac{1}{(a-x)^2} + \frac{1}{(a+x)^2} \right]$$

For U to be minimum, $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} > 0$,

$$\Rightarrow (a-x)^2 = (a+x)^2 \Rightarrow a+x = \pm(a-x)$$

$\Rightarrow x = 0$, because other solution is irrelevant.

Thus the charged particle at the origin will have minimum force and minimum P.E.

6. B

6. New electrostatic energy $U = \frac{kQq}{a} + \frac{kq^2}{a} + \frac{kQq}{a\sqrt{2}} = 0$

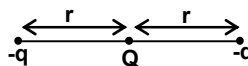
$$\frac{kq}{a} Q + q + \frac{Q}{\sqrt{2}} = 0 \Rightarrow Q = -\frac{2q}{2+\sqrt{2}}$$

7. $\frac{-3q^2}{8\pi\epsilon_0 r}$

7. $EPE = \frac{k(-q)(-q)}{2r}$

$$+ \frac{k(q)(-q)}{r} y^2$$

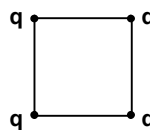
$$= -\frac{2kq^2}{r} + \frac{kq^2}{2r} = \frac{-3kq^2}{2r}$$



8. $\frac{4kq^2}{a} \left(1 + \frac{1}{2\sqrt{2}} \right)$

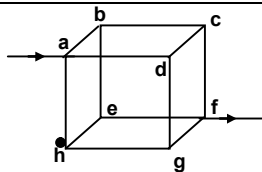
8. $EPE = 4 \times \frac{kq^2}{a}$

$$+ \frac{k\epsilon^2}{\sqrt{2}a} + 2$$



9. $\frac{4kq^2}{\sqrt{6}a} \left(3 + \frac{3}{\sqrt{2}} + \frac{1}{\sqrt{3}} \right)$

9. $\omega = u_f - u_i$
 $= 12 \times \frac{kq^2}{a} + 4 \times \frac{kq^2}{\sqrt{3}a}$
 $12 \times \frac{kq^2}{\sqrt{2}a}$



10. $r/4$

10. $u_f + k_i = u_i + k / f$

$$\frac{1}{2}mv^2 = \frac{KQq}{r} = v^2r = \text{const}$$

If speed is $2v$

$$= r' = \frac{v^2r}{(2v)^2} = \frac{r}{4}$$

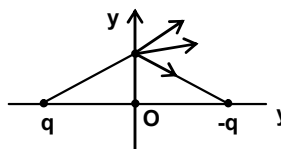


11. B

11. $V_0 = 0$

No work is done in moving a test charge from ρ to O

E field along x-axis toward left of q is -ve x-axis
dipole moment is along -ve x-axis.



Level-II

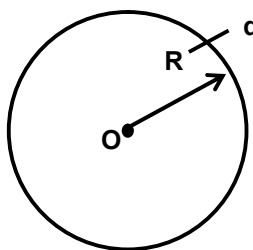
1. $\frac{3KQ^2}{8\pi R}$

1. $EPE = \frac{3KQ^2}{5R} + \int \frac{KQdq}{r}$

↑ ↑
Self energy intersection PE
of sphere

$$IPE = \int_0^R \frac{KQ}{r} \cdot 4\pi r^2 dr = \frac{3KQ^2}{R^3} \int_0^R r dr = \frac{3KQ^2}{2R}$$

$$EPE = \frac{3KQ^2}{4R} + \frac{3KQ^2}{2R} = \frac{21KQ^2}{10R}$$

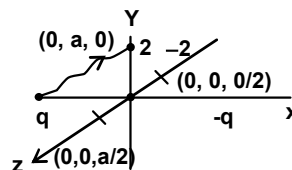


2. Zero

2. $U_1 = \frac{kq(1)}{\frac{\sqrt{5}}{2}a} + \frac{k(-q)(1)}{\frac{\sqrt{5}}{2} \frac{a}{2}} = 0$

$$U_2 = 0$$

$$W = 0$$

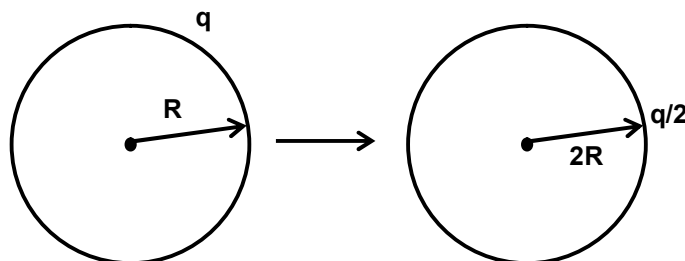


3. $\frac{7q^2}{64\pi\epsilon_0 R}$

3. $U_i = \frac{kq^2}{2R}$

$$U_f = \frac{k(q/2)^2}{2(2R)} = \frac{kq^2}{16R}$$

$$\omega_{EF} = U_i - U_f = \frac{kq^2}{2R} \left(1 - \frac{1}{8}\right) = \frac{7kq^2}{16R}$$



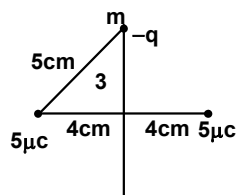
4. 10^3 m/sec

4. $U_i + K_i = U_f + K_f$

$$2 \times \frac{k(5 \times 10^{-6})(-q)}{5 \times 10^{-2}} = 2 \times \frac{k(5 \times 10^{-6})(-q)}{4 \times 10^{-2}} + \frac{1}{2}mv^2$$

$$\frac{1}{2}mv^2 = kq \times 10^{-3} \left(\frac{1}{4} - \frac{1}{5} \right) = kq \times 10^{-3} \times \frac{1}{20}$$

$$v^2 = \frac{q \times 10^{+5} \times 10^{-5}}{9 \times 10^{-6}} = 10^{+6}; \quad v = 10^3 \text{ m/sec}$$



5. $\frac{q^2}{4\pi\epsilon_0 a} (4 - 2\sqrt{2})$

5. $U_i = \frac{4k(q)(-q)}{a}$

$$+ \frac{kq^2}{\sqrt{2}a} \times 2$$

$$= \frac{\sqrt{2}kq^2}{a}$$

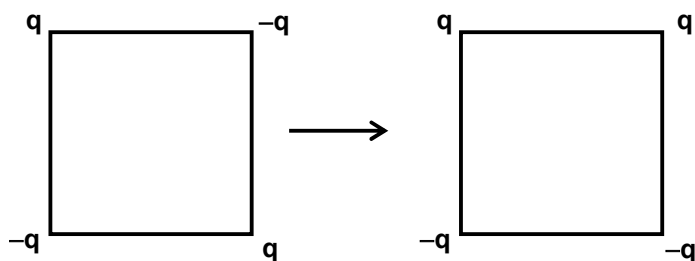
$$U_f = \frac{kq(-q)}{a} \times 2$$

$$\omega_{\text{ext}} = \Delta U = U_f - U_i$$

$$+ \frac{kq^2}{a} \times 2 - 1 \frac{k(q)(-q)}{\sqrt{2}a} \times 2$$

$$= \frac{4kq^2}{a} - \frac{2\sqrt{2}kq^2}{a}$$

$$= -\frac{\sqrt{2}kq^2}{a} = (4 - 2\sqrt{2}) \frac{kq^2}{a}$$



6. $6\sqrt{2} \text{ m}$

6. $u_i + k_i = u_f + k_f$

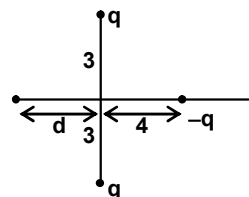
$$\frac{-kq^2}{5} \times 2 + 4 = \frac{-kq^2}{\sqrt{9+d^2}} \times 2$$

$$2kq^2 \left(\frac{1}{5} - \frac{1}{\sqrt{9+d^2}} \right) = 4$$

$$2 \times 9 \times 10^9 \times (5 \times 10^{-5})^2 \left[\frac{1}{5} - \frac{1}{\sqrt{9+d^2}} \right] = 4$$

$$45 \left[\frac{1}{5} - \frac{1}{\sqrt{9+d^2}} \right] = 4 = 9 - \frac{45}{\sqrt{9+d^2}} = 4 \Rightarrow \frac{45}{\sqrt{9+d^2}} = 5 \Rightarrow 0 = \sqrt{9+d^2} \Rightarrow d^2 = 72$$

$$D = 6\sqrt{2} \text{ m}$$



7. 2

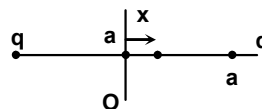
7. $U_f = \frac{2kQq}{a}$

$$U_i = \frac{kQq}{a+x} + \frac{kQq}{a-x}$$

$$= \frac{2kQqa}{a^2 - x^2} = \Delta U = U_f - U_i = 2kQq \left[\frac{a}{a^2 - x^2} - \frac{1}{a} \right]$$

$$= 2kQq \left(\frac{a^2 - a^2 + x^2}{a(a^2 - x^2)} \right)$$

$$\Rightarrow \Delta U = \frac{2kQq x^2}{a(a^2 - x^2)} \Rightarrow a = 2$$

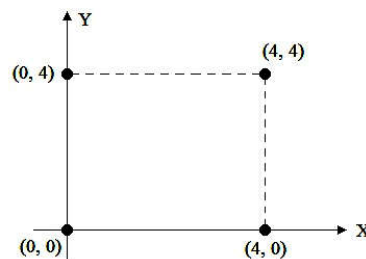


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ELECTROSTATICS -SHEET: 10(Lecture – 10)

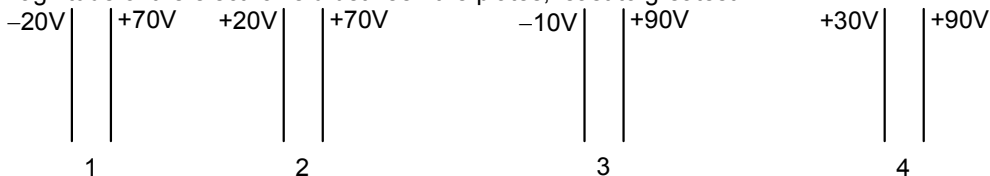
Level-I

- In an electric field the work done in moving a unit positive charge between two points is the measures of(Resistance / / Intensity of electric field)
- On moving a charge of 20 coulombs by 2 cm, 2J of work is done, then find the potential difference between the points.
- The electric field coming out of the equi-potential surface is-
(A) perpendicular to the surface (B) parallel to the surface
(C) in all directions (D) zero
- Find the work done If a unit charge is taken from one point to another over an equipotential surface.
- The electric field in a region of space has the components $E_y = E_z = 0$ and $E_x = (4.00 \text{ N/C})x$. Point A is on the y-axis at $y = 3.0 \text{ m}$ and point B is on the x-axis at $x = 4.0 \text{ m}$. What is the potential difference $V_B - V_A$?
- The kinetic energy of a charged particle decreases by 10 J as it moves from a point at potential 100 V to a point at potential 200 V. Calculate the charge on particle.
- A charge of $2 \times 10^{-7} \text{ C}$ moves from the co-ordinate origin to the point (4, 4) with co-ordinates measured in meters (see figure). Calculate the work done by the uniform electric field $E = (1.0 \times 10^4) \hat{i} \text{ N/C}$ for each of three paths:
(a) $(0, 0) \rightarrow (4, 0) \rightarrow (4, 4)$,
(b) $(0, 0) \rightarrow (4, 4)$, along the diagonal,
(c) $(0, 0) \rightarrow (0, 4) \rightarrow (4, 4)$,
(d) What is the electrostatic potential at point (4, 4) relative to the origin?



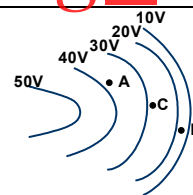
Q. 7

- The diagram shows four pairs of large parallel conducting plates. The plates are separated by equal distance in all cases. The value of the electric potential is given for each plate. Rank the pairs according to the magnitude of the electric field between the plates, least to greatest.

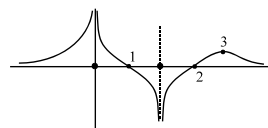


- The electric potential V at any point x, y, z in space is given by $V = 4x^2 \text{ volt / meter}^2$. The electric field at the point $(1 \text{ m}, 0, 2 \text{ m})$ is
- Electric field in a region is given by $\vec{E} = (2\hat{i} + 3\hat{j} - 4\hat{k}) \text{ V/m}$. Find the potential difference between points $(0, 0, 0)$ and $(1, 2, 3)$.
- Potential in the x - y plane is given as $V = 5(x^2 + xy) \text{ volts}$. Find the electric field at the point having coordinates $(1, -2, 0)$.

12. Given figure shows the lines of constant potential in a region in which an electric field is present. The values of potentials are as written. At which of the points A, B and C is the magnitude of the electric field greatest?

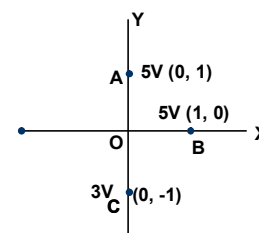
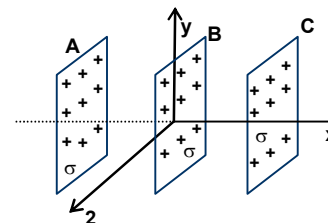


13. A point charge $+q$ and mass 10 gm experiences a force of 100 N at a point at a distance 20 cm from a long infinite uniformly charged wire. If it is released find its speed with it is at a distance 40 cm from wire.
14. Two points P and Q are maintained at the potentials of 10 V and -4 V, respectively. Calculate the work done in moving 100 electrons from P to Q.
15. Two point charges are kept at a certain distance from one another. The graph represents the variation of the potential along the straight line connecting the two charges. At what point is the electric field zero?



Level-II

1. Two points A and B are at distances of 'a' and 'b' respectively from an infinite conducting plate having charge density σ . Find the work done in moving charge Q from A to B.
2. Three infinite plane have a uniform surface charge distribution σ on its surface. All charges are fixed. On each of the three infinite planes, parallel to the y-z plane placed at $x = -a$, $x = 0$ and $x = a$, there is a uniform surface charge of the same density, σ . The potential difference between A and C.
3. Electrical potential in an electric field is given by $V = K/r$ (K being constant). If the position vector $\vec{r} = 2\hat{i} + 3\hat{j} + 6\hat{k}$, Calculate the electric field.
4. A uniform electric field exists in straight line in a certain region. The potentials at different points in the region are as shown in the figure. Find the magnitude and the direction of electric field in the region.



5. Find the work done by external agent in carrying a particle of a charge $2C$ from B (1 m, 0, 1 m) to A(0.8 m, 0.6 m 1 m) in a non-uniform electric field $E = (y\hat{i} + x\hat{j} + \hat{k})$ N/m along the straight line path from B to A.
6. Some equi-potential surfaces are shown in figures (1) and (2). What can you say about the magnitude and direction of the electric field?

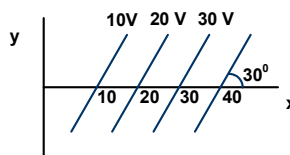


Figure 1

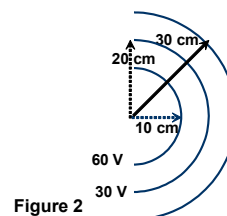
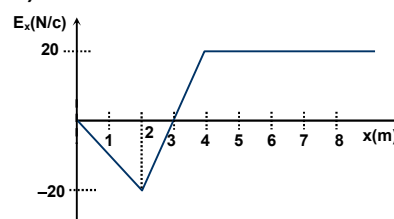


Figure 2

PARAGRAPH TYPE (7-8)

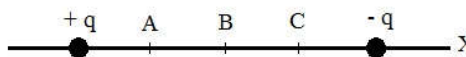
A graph of the x-component of the electric field as a function of x in a region of space is shown in the figure. The y and z-components of the electric field are zero in this region. The electric potential at the origin is 10 V.



7. The electric potential at $x = 2$ m is
(A) 10 V
(B) 20 V
(C) 30 V
(D) 40 V

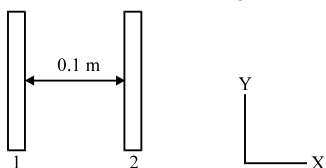
8. The greatest positive value of electric potential for points of the x-axis for which $0 \leq x \leq 6$ m is
 (A) 10 V (B) 20 V
 (C) 30 V (D) 40 V

9. Point charges $+q$ and $-q$ are located on the x-axis as drawn in figure. Three points (A, B, C) are indicated on the line joining the two charges. If $V_B = 0$ and V_A is 12 V higher than V_B and 24 V higher than V_C , determine V_A and V_C .



10. Two insulating plates are both uniformly charged in such a way that the potential difference between them is $V_2 - V_1 = 20$ V. (i.e. plate 2 is at a higher potential). The plates are separated by $d = 0.1$ m and can be treated as infinitely large. An electron is released from rest on the inner surface of plate 1. What is its speed when it hits plate 2?

($e = 1.6 \times 10^{-19}$ C, $m_e = 9.11 \times 10^{-31}$ kg) –



11. The potential at a point x (measured in μm) due to some charges situated on the x-axis is given by $V(x) = 20/(x^2 - 4)$ volts. Find the electric field E at $x = 4 \mu\text{m}$.

CPP

ELECTROSTATICS -SHEET: 8(Lecture – 8)

ANSWER KEYLevel-I

1. Potential difference

2. 0.1 V

2. $\theta = 20^\circ$ $S = 2\text{cm}$

$$W = \vec{F} \cdot \vec{S} = Q\vec{E} \cdot \vec{S} = Q\Delta V$$

$$W = 2\text{J} \Rightarrow \Delta V = \frac{2\text{J}}{20\text{C}} = 0.1\text{volts}$$

3. A

4. Zero

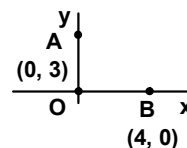
5. -32V

$$E_x = 4x \frac{\text{N}}{\text{C}}$$

$$E_y = E_z = 0$$

$$V_B - V_A = -\int_A^B \vec{E} \cdot d\vec{\ell}$$

$$= -\int_A^O E \cdot d\ell - \int_O^B E \cdot d\ell = -\int_0^4 4x \, dx = 32\text{V}$$



6. 0.1C

$$w = \Delta KE = -Q\Delta V \Rightarrow \theta = \frac{\Delta KE}{-\Delta V} = \frac{-10}{-(200-100)} = +0.1\text{c}$$

7. Work done is same in all paths = -8mJ d) $-4 \times 10^4\text{V}$

$$\vec{E} = 10^4 \hat{i} \frac{\text{N}}{\text{C}}; Q = 2 \times 10^{-1}\text{C}$$

$$w_1 = 2 \times 10^{-7} \times 10^4 \times 4 + 0$$

$$= 8 \times 10^{-3}\text{J}$$

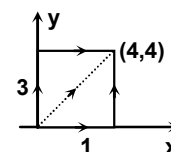
$$W_2 = 2 \times 10^{-7} \times 10^4 \times 4 + 0$$

$$= 8 \times 10^{-3}\text{J}$$

$$W_3 = 2 \times 10^{-7} \times 10^4 \times 4\sqrt{2} \times \frac{1}{\sqrt{2}} = 8 \times 10^{-3}\text{J}$$

8 mJ Ans

$$(d) \Delta V = -4 \times 10^4\text{V}$$



8. 2, 4, 1, 3

	(I)	(II)	(III)	(IV)
8. $E = -\left(\frac{\Delta V}{\Delta \ell}\right)$	$+\frac{10}{d}$	$+\frac{50}{d}$	$+\frac{100}{d}$	$+\frac{60}{d}$

9. $-8\hat{i}$

$$V = 4x^2$$

$$E_x = -\frac{\partial V}{\partial x} = -8x \text{ at pt } (1, 0, 2)$$

$$E_4 = E_2 = 0 \quad \vec{E} = -8\hat{i}$$

10. P.d. across the points = $-\vec{E} \cdot \Delta \vec{r}$

$$V_2 - V_1 = -(2\hat{i} + 3\hat{j} - 4\hat{k}) \cdot (\hat{i} + 2\hat{j} + 3\hat{k}) = -2 - 6 + 12 = 4\text{ volts.}$$

10. $\vec{E} = 2\hat{i} + 3\hat{j} - 4\hat{k}$

$A : (0, 0, 0) B : (1, 2, 3)$

$$\Delta V = -\int \vec{E} \cdot d\vec{v} = -\int (2dx + 3dy - 4dz)$$

$$= -2\Delta x - 3\Delta y + 4\Delta z \Big|_{(0,0,0)}^{(1,2,3)}$$

$$= 12 - 6 - 2 = 4V$$

11. $-5\hat{j} \text{ V/m}$

11. $V = 5(x^2 + xy) \quad \vec{E} = -\frac{\partial V}{\partial x}\hat{i} - \frac{\partial V}{\partial y}\hat{j} - \frac{\partial V}{\partial z}\hat{k}$

$$\vec{E}(1, -2, 0) = (-10x - 5y)\hat{i} - 5x\hat{j}$$

$$= -5\hat{j}$$

12. $\vec{E} = -\frac{dv}{dr}\hat{n}$

The potential difference between any two consecutive lines is $dV = V_1 - V_2 = 10 \text{ V}$
= constant

Hence, E will be maximum where the distance dr between the lines is minimum, i.e. at B where the lines are closest.

13. $20\sqrt{10\ln 2}$

13. $F = 100N = \left(\frac{2k\lambda}{r}\right)q$

at $r = 20 \text{ cm}$

$$100 = \frac{2k\lambda q}{2} \quad k\lambda q = 10$$

$$a = \frac{2k\lambda q}{m r} = \frac{20}{0.1r} = \frac{2000}{r}$$

$$\frac{v dv}{dr} = \frac{2000}{r} \quad 31 \int_0^v v dv = 2000 \int_{20}^{40} \frac{dr}{r}$$

$$\frac{v^2}{2} = 2000 \ln 2 \quad av = 20\sqrt{10\ln 2}$$

14. $2.24 \times 10^{-16} \text{ J}$

14. $q = 100e^-$

$$\omega = \epsilon \Delta V$$

$$= 100 \times 1.6 \times 10^{-19} \times 14$$

$$= 2.24 \times 10^{-16} \text{ J}$$

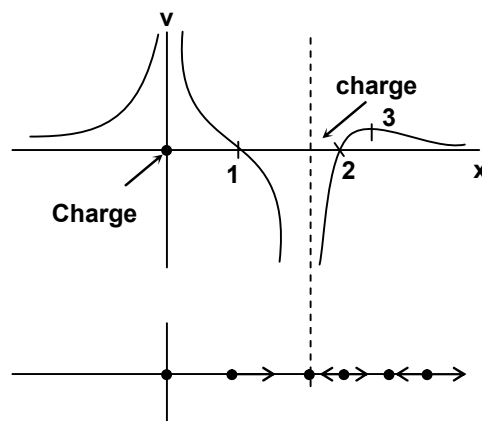
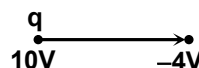
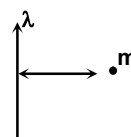
15. 3

15. Field is zero at (3)

$$E = -\frac{du}{dx} \text{ at}$$

$$(3) \frac{du}{dx} = 0$$

$$\Rightarrow E = 0$$

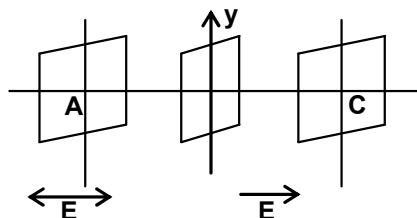


Level-II

$$1. \quad E = \frac{\sigma}{\epsilon_0}$$

$$\therefore W = \frac{Q\sigma}{\epsilon_0}(b-a)$$

2. Zero
 2. A to C
 $\Rightarrow \Delta V = 0$



$$3. \quad \frac{(2\hat{i} + 3\hat{j} + 6\hat{k})}{343} K$$

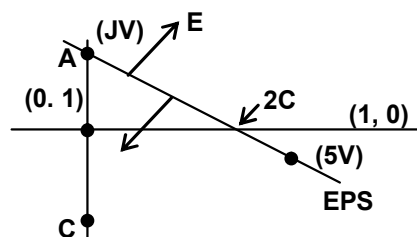
$$3. \quad V = \frac{k}{r}$$

$$E = \frac{dV}{dr} = \frac{K}{r^2} (\vec{E}) = \frac{k}{|\vec{r}|^2} = \frac{k}{49} \vec{E} = |\vec{E}| \hat{r} = \frac{k}{49} \left(\frac{2\hat{i} + 3\hat{j} + 6\hat{k}}{7} \right)$$

4. $\sqrt{2} \text{ V/m.}, \vec{E} = -\hat{i} - \hat{j}$

4. Field is either in dir. $\hat{i} + \hat{j}$
 Field is either in dir. $\hat{i} - \hat{j}$
 field is in dir. of dic. Potential

$$x \hat{E} = \frac{(-\hat{i} - \hat{j})}{\sqrt{2}} |E| = \frac{\Delta V}{\ell} = \frac{2}{\sqrt{2}} = \sqrt{2}$$



5. -0.96 J

5. 2C from (1, 0) to (0.8, 0.6, 1)
 $\vec{E} = (y\hat{i} + x\hat{j} + \hat{k})$
 $W = q\Delta V = -q \int E \cdot d\vec{l} = -q \int (y dx + x dy + dz)$
 $= -q \int [dx y + \int dz] = -q [xy + z]_{(1,0,1)}^{(0.8,0.6,1)}$
 $= -2 [(0.48 + 1) - (1)]$
 $= 0.96 \text{ J}$

6. Electric field is perpendicular to the equipotential surface and in the direction of decreasing potential.
 Figure 1:
 The electric field will be making an angle of 120° with x-axis.

$$\text{Magnitude of electric field along x-axis} = E \cos 120^\circ = -\frac{(20-10)}{(20-10) \times 10^{-2}}$$

$$-E \cdot \frac{1}{2} = -\frac{10}{0.10} \Rightarrow E = 200 \text{ V/m}$$

Figure 2:

Direction of electric field will be radially outward, $rV = \text{constant} = 6 \text{ (Vm)}$

$$\therefore \text{Potential at any distance } r \text{ from the center, } V(r) = \frac{6}{r}$$

$$\text{Hence, } E = -\frac{dV}{dr} = \left(\frac{6}{r^2} \right) \text{ V/m}$$

7. C

$$7. \quad \Delta V = -\int E \cdot dt = -\left[\frac{1}{2} \times 2 \times .20\right] = 20$$

$$V(x=2) - V(x=0) = 20$$

$$V(x=2) = 30 \text{ V(C)}$$

8. D

8. For greatest V it will be at $x = 3$

$$V(x=3) = V(x=0) + \frac{1}{2} \times 3 \times (+20) = 40 \text{ V}$$

$$9. \quad V_A = -V_C = 12 \text{ V}$$

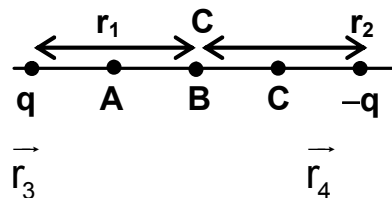
$$9. \quad \text{since } V_0 = 0 \Rightarrow r_1 = r_2 = r$$

$$V_A - V_B = 12$$

$$V_A - V_C = 24 = V_C = -12$$

$$V_A = \frac{kq}{r_3} - \frac{kq}{(2r - r_3)} = 12$$

$$\frac{kq}{2r - r_4} - \frac{kq}{r_4} = -12$$



$$10. \quad 2.65 \times 10^6 \text{ m/s}$$

$$10. \quad \Delta V = 20 \text{ V} = V_2 - V_1$$

$$R\Delta V = \Delta KE = \frac{1}{2} m V^2$$

$$\Rightarrow V^2 = \sqrt{\frac{2e\Delta V}{m}}$$

$$2\sqrt{\frac{2 \times 1.6 \times 10^{-19} \times 20}{9.1 \times 10^{-31}}}$$

$$\sqrt{7.03 \times 10^{12}}$$

$$V = 2.65 \times 10^6 \text{ m/s}$$

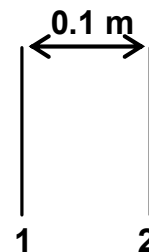
11. $10/9 \text{ Volt}/\mu\text{m}$ and in the +ve x direction

$$11. \quad V(x) = \frac{20}{x^2 - 4}$$

$$E_x = -\frac{dv}{dx} = \left(\frac{-20(2x)}{(x^2 - 4)^2} \right)$$

$$E_x = \frac{40x}{(x^2 - 4)^2} \text{ at } x = 4 \mu\text{m}$$

$$= \frac{40 \times 4}{12 \times 12} = \frac{10}{9} \text{ V}/\mu\text{m}$$

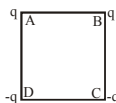


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ELECTROSTATICS -SHEET: 11 (SHEET-11)

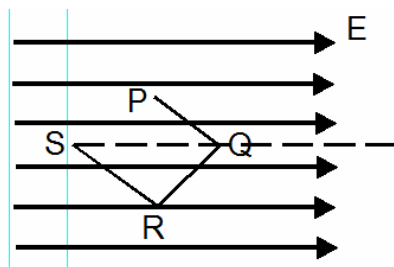
Level-I

- If you know $\vec{E}$ at a given point, can you calculate V at that point? If not, what further information do you need.
 - If $\vec{E}$ equals zero at a given point, must V equal zero for that point?
- Eight small drops, each of radius r and having same charge q are combined to form a big drop. The ratio between the potentials of the bigger drop and the smaller drop is
(A) 8 : 1 (B) 4 : 1 (C) 2 : 1 (D) 1 : 8
- Two small identical metal balls A and B of radius ' r ' are placed apart. The distance between centre of balls is ' a_0 '. The net potential of ball A is V_1 and that of B is V_2 . Let q_1 and q_2 are the charges on balls A and B respectively. Then the charge on A and B are (given $r \ll a_0$)
 (A) $q_1 = 4\pi\epsilon_0 \frac{a_0 r (V_1 a_0 - V_2 r)}{a_0^2 - r^2}$ (B) $q_1 = 4\pi\epsilon_0 \frac{a_0 r (V_1 a_0 + V_2 r)}{a_0^2 + r^2}$
 (C) $q_2 = 4\pi\epsilon_0 \frac{a_0 r (V_2 a_0 + V_1 r)}{a_0^2 + r^2}$ (D) $q_2 = 4\pi\epsilon_0 \frac{a_0 r (V_2 a_0 - V_1 r)}{a_0^2 - r^2}$
- Two fixed charges $-2Q$ and Q are located at points with coordinates $(-3a, 0)$ and $(+3a, 0)$ respectively in the $x - y$ plane. Find the locus in the $x - y$ plane where the electric potential due to the system is zero.
- Three charges $2q$, $-q$ and $-q$ are located at the vertices of an equilateral triangle. At the centre of the triangle:
(A) the field is zero but potential is non-zero (B) the field is non-zero but potential is zero
(C) both the field and potential are zero (D) both field and potential are non-zero.
- A hollow metal sphere of radius 5cm is charged such that the potential on its surface is 10V. Calculate the potential at the centre of the sphere.
- A solid conducting sphere having a charge Q is surrounded by an uncharged concentric conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V . If the shell is now given a charge of $3Q$, find the new potential difference between the same two surfaces
- Plot the dependence of electric potential V on the distance ' r ' from the centre of a charged spherical shell.
- Charges are placed on the vertices of a square as shown. Let E be the electric field and V the potential at the centre. If the charges on A and B are interchanged with those on D and C respectively, then



- (A) E remains unchanged, V changes (B) Both E and V change
(C) E and V remain unchanged (D) E changes, V remains unchanged

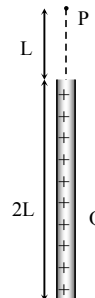
- Point charge (q) moves from point (P) to point (S) along the path PQRS as shown in fig. in a uniform electric field E pointing coparallel to the positive direction of the x -axis. The co-ordinates of the points P, Q, R and S are $(a, b, 0)$, $(2a, 0, 0)$, $(a, -b, 0)$ and $(0, 0, 0)$ respectively. The work done by the field in the above process is given by the expression



11. Two identical thin rings, each of radius R metres, are coaxially placed at a distance (R) metres apart. If Q_1 coul and Q_2 coul are respectively The charges uniformly spread on the two rings. The work done in moving a charge (q) from the centre of one ring to that of other is -
12. Two spherical conductors A and B of radii 1 mm and 2 mm are separated by a distance of 5 cm and are uniformly charged. If the spheres are connected by a conducting wire. In equilibrium condition, calculate the ratio of the magnitude of the electric fields at the surfaces of spheres A and B.

Level-II

1. A point charge q is placed at the centre of hollow conducting sphere of inner radius R and outer radius $2R$. Work done required to slowly remove the point charge from centre to infinity (through a small orifice) is $\frac{1}{n} \frac{Q^2}{4\pi\epsilon_0 R}$ find n .
2. A point charge $-q$ is placed at centre of a hollow conducting sphere with radius R carrying a charge $+Q$. Calculate the potential of a point in a space at a distance of $r > R$ from the centre of the sphere.
3. Find a charge Q is uniformly distributed on a dielectric rod AB of length $2L$. The potential of P shown in figure.



4. A nonconducting ring of radius 0.5 m carries a total charge of 1.11×10^{-10} C distributed non-uniformly on its circumference producing an electric field E everywhere in space. Evaluate the value of the line integral $\int_{l=\infty}^{l=0} \vec{E} \cdot d\vec{l}$. ($l = 0$ being centre of the ring) in volts.
5. A charge $+q$ is fixed at each of the point $x = x_0, x = 3x_0, x = 5x_0, \dots$ upto infinity on the x -axis, and charges $-q$ is fixed at each of the point $x = 2x_0, x = 4x_0, x = 6x_0, \dots$ upto infinity. Here x_0 is a positive constant. Find the potential at the origin due to the above system of charges.
6. Consider a system of three charges $q/3, q/3$ and $-2q/3$ placed at point A, B and C, respectively, as shown in the figure. Take O to be the centre of the circle of radius R and angle $CAB = 60^\circ$.

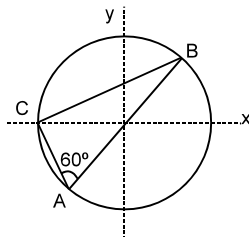


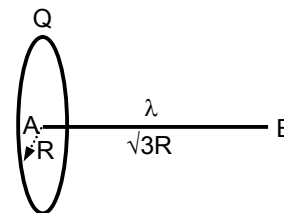
Figure :

- (A) The electric field at point O is $\frac{q}{8\pi\epsilon_0 R^2}$ direction along the negative x -axis
- (B) The potential energy of the system is zero
- (C) The magnitude of the force between the charges at C and B is $\frac{q^2}{54\pi\epsilon_0 R^2}$

(D) The potential at point O is $\frac{q}{12\pi\epsilon_0 R}$

7. A conducting sphere of radius 10cm is charged 10 μC . Another uncharged sphere of radius 10 cm is allowed to touch it for some time. After that if the sphere are separated, what will be the ratio of the surface charge density on the spheres.

8. A non conducting rod AB of length $\sqrt{3}R$, uniformly distributed charge of linear charge density λ and a non-conducting ring of uniformly distributed charge Q, are placed as shown in the figure. Point A is the centre of ring and line AB is the axis of the ring, perpendicular to plane of ring. Calculate the electrostatic interaction energy between ring and rod.



9. An electric charge $10^{-3} \mu\text{C}$ is placed at the origin (0, 0) of X - Y co-ordinate system. Two points A and B are situated at $(\sqrt{2}, \sqrt{2})$ and (2, 0) respectively. Find the potential difference between the points A and B.
10. An electron is released from rest at one point in a uniform electric field and travels a distance of 10 cm in 10^{-7} seconds. Calculate potential difference across the points.

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ELECTROSTATICS -SHEET: 11(PRACTICE SHEET-1)

ANSWER KEY

Level-I

1. (i) No, we need the information about nature of $\vec{E}$ even if we take reference at infinity.
(ii) No

2. (B) By using $V_{\text{big}} = n^{2/3} V_{\text{small}}$ $\therefore \frac{V_{\text{big}}}{V_{\text{small}}} = (8)^{2/3} = \frac{4}{1}$

3. A, D

3. $V_1 = \frac{kq_1}{r} + \frac{kq_2}{a_0}$... (i)

$$V_2 = \frac{kq_2}{r} + \frac{kq_1}{a_0} \quad \dots (ii)$$

On solving get the result.

4. A circle of radius 4a and centre at (5a, 0)

$$4. \quad \frac{kQ}{(x-3a)^2 + y^2} - \frac{kqQ}{\sqrt{(x+3a)^2 + y^2}} = 0$$

$$\frac{1}{\sqrt{(x-3a)^2 + y^2}} = \frac{2}{\sqrt{(x+3a)^2 + y^2}}$$

$$4[(x-3a)^2 + y^2] = (x+3a)^2 + y^2$$

$$4(x-3a)^2 + 4y^2 = (x+3a)^2 + y^2$$

$$4[x^2 + 9a^2 - 6xa] + 4y^2 = x^2 + 9a^2 + 6xa + y^2$$

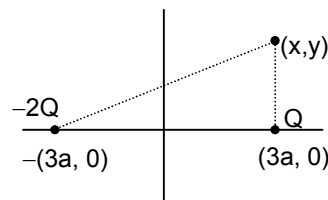
$$3x^2 + 3 \times 9a^2 - 30xa + 3y^2 = 0$$

$$x^2 + y^2 + 9a^2 - 10xa = 0$$

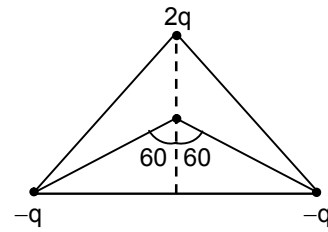
$$(x-5a)^2 + y^2 - a^2 = 0$$

$$(x-5a)^2 + y^2 = a^2$$

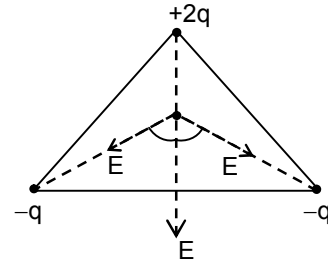
Above is circle with centre (5a, 0) and radiuses.



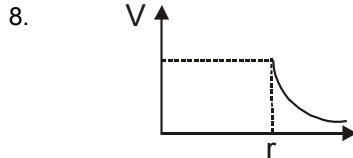
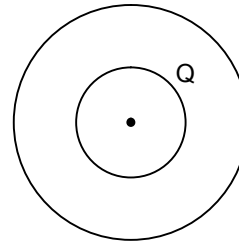
5. B
 5. $2q, -q, q$
 Conceptual
 Its clear that electric field is non zero.



6. 10V
 6. Potential inside sphere is same as that on the surface

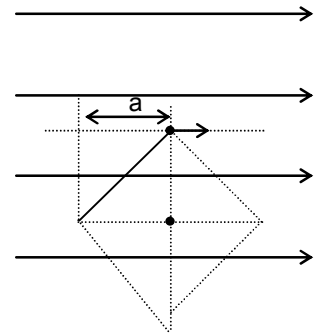


7. V
 7. the potential difference between



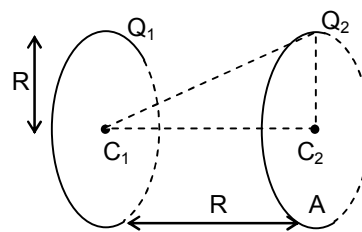
9. D

10. $-q E a$
 10. $w = (q E a \text{ ed } 180)$
 $= -q E a^2$



11.
$$\frac{q(Q_1 - Q_2)(\sqrt{2} - 1)}{(\sqrt{2.4} \epsilon_0 \pi R)}$$

$$\begin{aligned}
 11. \quad V_1 &= \frac{kQ_1}{R} + \frac{kQ_2}{\sqrt{2}R}, \quad V_2 = \frac{kQ_2}{R} + \frac{kQ_1}{\sqrt{2}R} \\
 w &= (V_2 - V_1) \\
 w &= \frac{kQ_2}{R} + \frac{kQ_1}{\sqrt{2}R} - \frac{kQ_1}{R} - \frac{kQ_2}{\sqrt{2}R} \\
 &= (Q_1 - Q_2) \frac{k}{\sqrt{2}R} \frac{k}{R} [Q_2 - Q_1] \\
 &= \frac{(Q_1 - Q_2)k}{\sqrt{2}R} - \frac{k}{R} (Q_1 - Q_2) \\
 &= \frac{Q_1 - Q_2}{R} \cdot k \left[\frac{1}{\sqrt{2}} - 1 \right]
 \end{aligned}$$



12. 2 : 1

12. $\frac{kq_1}{1} = \frac{kq_2}{2}$

$$q_1 = \frac{q_2}{2}$$

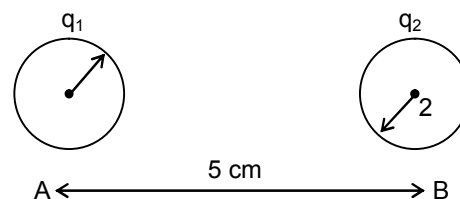
$$E_1 = \frac{kq_1}{1^2}$$

$$E_1 = \frac{kq_1}{2^2}$$

$$\frac{E_1}{E_2} = \frac{q_1}{q_2} \times \left(\frac{2}{1}\right)^2$$

$$= \begin{pmatrix} 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 4 \\ 1 \end{pmatrix} = 2$$

(1)



Level-II

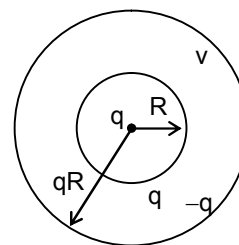
1. 4

1. $w = (v_f - v_i)$

$$v_i = \frac{kq^2}{2R} + \frac{kq^2}{2(2R)} - \frac{kqq}{2R} + \frac{kq^2}{R} - \frac{kq^2}{2R}$$

$$= \frac{kq^2}{4R} + \frac{kq^2}{2R} = \frac{kq^2}{2R} \left[\frac{3}{2} \right]$$

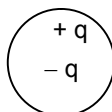
$$= \frac{3kq^2}{4R}$$



$$2. \quad \frac{Q - q}{4\pi\epsilon_0 r}$$

$$2. \quad v = \frac{k(Q-q)}{r}$$

$(d - q)$



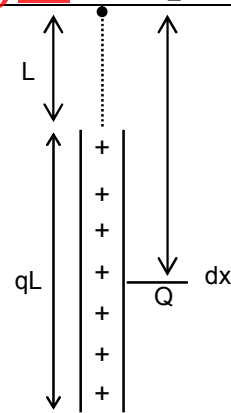
$$3. \quad \frac{Q \ln 3}{4\pi\epsilon_0(2L)}$$

3. $dq = \left(\frac{Q}{2L}\right) dx$

$$dq = \frac{k dq}{x}$$

$$dr = k \left(\frac{Q}{2L}\right) \int_L^{2L} \frac{dx}{x}$$

$$= \frac{kQ}{2L} \ln 2$$



4. 2

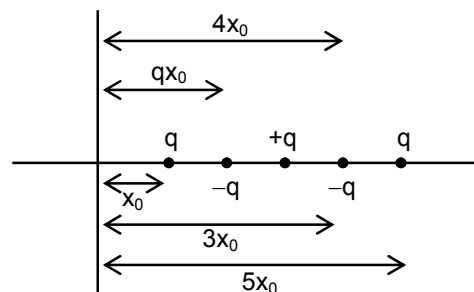
4. $v \sec v = \frac{kq}{R}$

5. $\frac{q \ln 2}{4\pi \epsilon_0 x_0}$

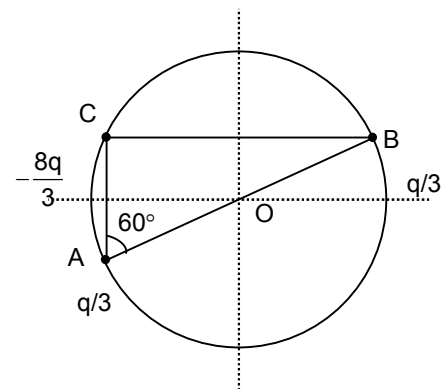
5. $v = \frac{kq}{x_0} - \frac{kq}{qx_0} + \frac{kq}{3x_0} - \frac{kq}{4x_0} + \dots$

$$v = \frac{kq}{x_0} \left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \right]$$

$$= \frac{kq}{x_0} \ln 2$$



6. C



7. 1:1

7. After redistribution new charges on spheres are

$$Q'_1 = \frac{10}{10+20} \cdot 10 = \frac{10}{3} \text{ mC}$$

$$\text{and } Q'_2 = \frac{20}{10+20} \cdot 10 = \frac{20}{3} \text{ mC}$$

$$\text{Ratio of charge densities } \frac{s_1}{s_2} = \frac{Q_1}{Q_2} \cdot \frac{r_2^2}{r_1^2}$$

$$= \frac{10/3}{20/3} \cdot \frac{20^2}{10^2} = \frac{2}{1}$$

$$\frac{1}{1} s = \frac{Q}{4\pi r^2}$$

8. $\frac{Q\lambda}{4\pi\epsilon_0} \ln(2+\sqrt{3})$

$$dU = V_{P(\text{ring})} \lambda dx = \frac{kQ\lambda}{\sqrt{R^2 + x^2}} dx$$

$$U = \frac{Q\lambda}{4\pi\epsilon_0} \int_0^{\sqrt{3}R} \frac{dx}{\sqrt{R^2 + x^2}}$$

$$\text{Let } x = R \tan \theta$$

$$\Rightarrow dx = R \sec^2 \theta d\theta$$

$$\text{If } x = 0 \Rightarrow \theta = 0, \text{ and } x = \sqrt{3}R$$

$$\Rightarrow \theta = \pi/3$$

9. Zero

$$9. \quad v_1 = \frac{kq}{2}$$

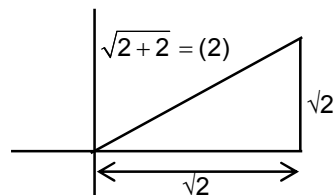
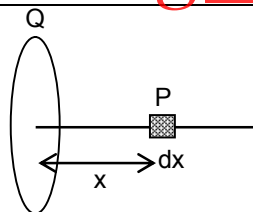
$$v_2 = \frac{kq}{2}$$

$$\Delta V = 0$$

10. $V = Es$;

$$E = \frac{2sm}{qt^2}$$

$$\Rightarrow V = \frac{2s^2m}{qt^2} = 11.375 \text{ Volt}$$



CPP

ELECTROSTATICS -SHEET: 12(Lecture – 12)

Level-I

1. Two concentric conducting spheres of radii R and $3R$ carrying charges Q and $2Q$ respectively. If the charge on inner sphere is doubled. Calculate the potential difference between inner and outer spheres.

2. A solid metallic sphere is placed in a uniform electric field. Which of the curves shown in the figure 298 represent the lines of force correctly?

(A) (a)
(C) (c)

(B) (b)
(D) (d)

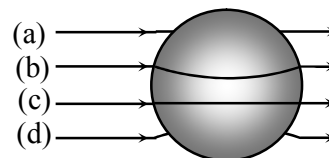


Figure 3

3. An ellipsoidal cavity is carved within a perfect conductor (see figure). A positive charge q is placed at the centre of the cavity. The points A and B are on the cavity surface as shown in figure-296. Then:

(A) electric field near A in the cavity = electric field near B in the cavity
(B) charge density at A = charge density at B
(C) potential at A = potential at B
(D) total electric flux through the surface of the cavity is q/ϵ_0

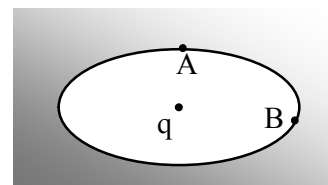


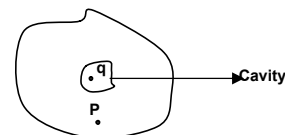
Figure 1

4. Two spherical conductors A and B of radii R_A and R_B ($R_A > R_B$) are charged and then connected by a conducting wire:

(A) both A and B have same charge
(B) both A and B have same charge and are at equal potential finally
(C) charge on A is zero and all charges appear on B and both are at equal potential finally
(D) surface density of A is equal to that of B and both are at equal potential finally

5. Inside an electrical conductor there is a cavity and a charge q is suspended fixed inside the cavity space without any electrical contact with conductor. Let $\vec{E}_1$ be the electrostatic field due to charge q at P. $\vec{E}_2$ be the electrostatic field at P due to induced charge on the inner surface of cavity. $\vec{E}_3$ be the electrostatic field at P due to the induced charge on the outer surface of conductor then

(A) $\vec{E}_1 + \vec{E}_2 + \vec{E}_3 = 0$
(B) $\vec{E}_1 + \vec{E}_2 = 0$
(C) $\vec{E}_1 + \vec{E}_3 = 0$
(D) $\vec{E}_3 = 0$



6. There is a spherical conductor of radius a . A point charge is placed at a distance $4a$ from the centre of the spherical conductor as shown in figure.

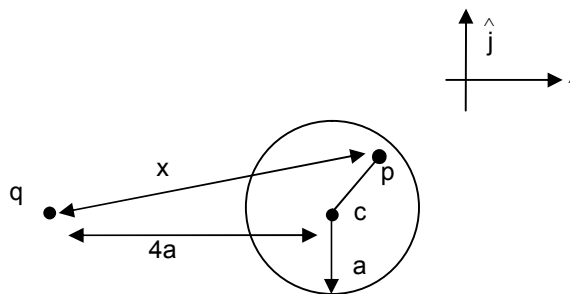
(A) The electric field at 'c' due to spherical conductor is

$$\frac{kq}{16a^2} \left(-\hat{i} \right).$$

(B) Potential at P due to induced charge on spherical conductor is $\left(\frac{kq}{4a} - \frac{kq}{x} \right)$.

(C) Net electric field at 'P' is ZERO.

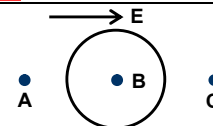
(D) Now the conductor is earthed then charge on the spherical conductor is $\left(-\frac{q}{4} \right)$



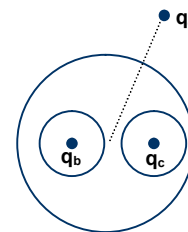
7. Two conducting spheres of unequal radii are given charge such that they have the same charge density. If they are now brought in contact,

(A) No charge will be exchanged between them
(B) Charge will flow from the larger to the smaller sphere
(C) Charge will flow from the smaller to the larger sphere
(D) No heat will be produced

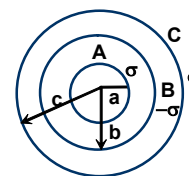
8. In a uniform electric field E , a dielectric (dielectric constant $\neq 1$) in the form of a sphere is introduced. How will the intensity of the field at points A, B and C change?
- (A) Field at A will increase.
 (B) Field at B will decrease.
 (C) Field at C will increase.
 (D) Field at all points will remain same.



9. A spherical conductor A contains two spherical cavities. The total charge on the conductor itself is zero. However, there is a point charge q_b at the centre of one cavity and q_c at the centre of the other. A considerable distance r away from the centre of the spherical conductor, there is another charge q_d . Forces acting on q_b , q_c and q_d are F_1 , F_2 and F_3 respectively. [Assume all charges are positive]
- (A) $F_1 < F_2 < F_3$ (B) $F_1 = F_2 < F_3$
 (C) $F_1 = F_2 > F_3$ (D) $F_1 > F_2 > F_3$



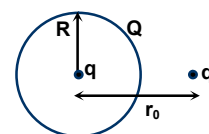
10. The adjacent figure shows charged spherical shells A, B and C having charge densities σ , $-\sigma$, σ and radii a , b , c , respectively. If $V_A = V_C$, then find b in terms of a and c .



Level-II

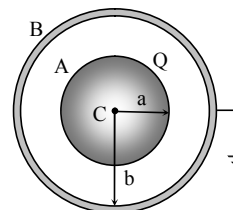
1. There are four concentric shell A, B, C and D of radii a , $2a$, $3a$, and $4a$ respectively. Shells B and D are given charges $+q$ and $-q$ respectively. Shell C is now earthed. Find the potential difference $V_A - V_C$.

2. A thin, metallic spherical shell contains a charge Q on it. A point charge q is placed at the centre of the shell and another charge q_0 is placed outside it at distance r_0 , as shown in figure. All three charges are positive. Find (a) magnitude of force on q due to charge q_0 , (b) magnitude of net force on charge q due to all charges in space

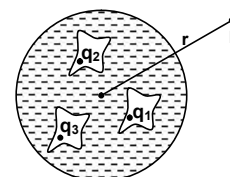


3. Inducting sphere A of radius a , with charge Q , is placed concentrically inside a conducting shell B of radius b . B is earthed. C is the common centre of A and B. Find

- (a) The field at a distance r from C, where $a \leq r \leq b$.
 (b) The potential difference between A and B.
 (c) The potential at the distance r from C, where $a \leq r \leq b$.



4. A conducting material whose outer surface is spherical and inside which there are three cavities as shown in figure. If charges q_1 , q_2 and q_3 are placed inside the cavities, find the potential at point P which is at distance r from the centre of sphere.



5. Three concentric spherical metallic shells A, B, and C of radii a , b , and c ($a < b < c$) have charge densities of σ , $-\sigma$ and σ respectively, then match the following:

Column I	Column II
(A) The potential at A	(p) $\frac{\sigma}{\epsilon_0} \left(\frac{a^2}{c} - \frac{b^2}{c} + c \right)$
(B) The potential at B	(q) $\frac{\sigma}{\epsilon_0} \left(\frac{a^2}{b} - b + c \right)$
(C) The potential at C	(r) $\frac{\sigma}{\epsilon_0}$

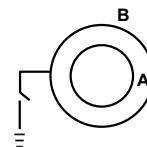
(D) The electric field at surface of A

(s) $\frac{\sigma}{\epsilon_0}(a - b + c)$

6. Initially the sphere A and B are at potentials V_A and V_B . The potential of A when sphere B is earthed is

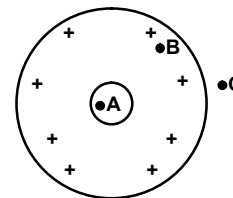
(A) zero
(C) $V_A - V_B$

(B) V_A
(D) none of these.

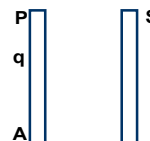


7. A conductor given a positive charge have a cavity. Then field strengths E_A , E_B and E_C at point A (within cavity), at B (within conductor but outside cavity) and C (outside conductor) will be

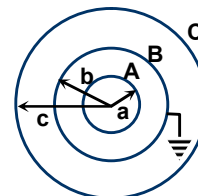
(A) $E_A = 0$, $E_B = 0$, $E_C = 0$
(B) $E_A = 0$, $E_B = 0$, $E_C \neq 0$
(C) $E_A \neq 0$, $E_B = 0$, $E_C \neq 0$
(D) $E_A \neq 0$, $E_B \neq 0$, $E_C \neq 0$



8. Two conducting plates P and S with large surface area A are placed as shown in the figure. A charge q is given to plate P. Determine the electric field between the plates at any point.



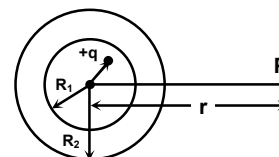
9. Figure shows three concentric thin spherical shells A, B and C of radii a, b and c, respectively. The shells A and C are given charge q and $-q$, respectively, and shell B is earthed. Find the charges appearing on the surfaces of A, B and C.



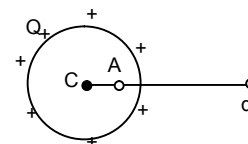
10. A thin spherical conducting shell of radius R has a charge q. Another charge Q is placed at the centre of the shell. The electrostatic potential at a point P a distance R/2 from the centre of the shell is –
11. Three concentric metallic spherical shells of radii R, 2R, 3R are given charges Q_1 , Q_2 , Q_3 respectively. It is found that the surface charge densities on the outer surfaces of the shells are equal. Find the ratio of the charges given to the shells, $Q_1 : Q_2 : Q_3$.

12. A conducting shell having inner radius R_1 and outer radius R_2 contains a charge +q which is placed at a distance x from its centre. Field at an exterior point P which is situated at a distance r from centre of shell ($r > R_2$)

(A) depends on x
(B) depends on R_1 and R_2
(C) depends only on r
(D) none



13. Consider a metallic sphere (solid) of radius R carrying charge Q on it. Another point charge 'q' is kept at P at a distance 3R from centre of the sphere. Find electrical potential at point A lying on the line joining C and P at distance R/2 from C.



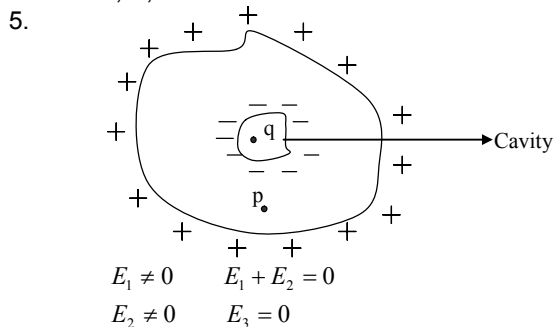
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ELECTROSTATICS -SHEET: 12(Lecture – 12)

ANSWER KEYLevel-I

1. becomes two times
 2. D 3. C 4. D

5. A, B, D



6. A, B, C, D

7. B

7. Let the spheres have radii r_1 and r_2 ($r_1 > r_2$).

$$V_1 = \frac{4\pi r_1^2 \sigma}{4\pi \epsilon_0 r_1} = \frac{r_1 \sigma}{\epsilon_0}, V_2 = \frac{4\pi r_2^2 \sigma}{4\pi \epsilon_0 r_2} = \frac{r_2 \sigma}{\epsilon_0}$$

$$\therefore V_1 > V_2$$

$\therefore$ charge will flow from the larger to the smaller sphere.

8. B 9. B

10. c-a

Level-II

1. $Kq/6a$
 2. (a) $\frac{1}{4\pi \epsilon_0} \frac{qq_0}{r_0^2}$, (b) 0

3. (a) $k \frac{Q}{r^2}$ (b) $kQ \left(\frac{1}{a} - \frac{1}{b} \right)$ (c) $kQ \left(\frac{1}{r} - \frac{1}{b} \right)$

4. $\frac{1}{4\pi \epsilon_0} \left(\frac{q_1 + q_2 + q_3}{r} \right)$

5. (A) $\rightarrow$ (s), (B) $\rightarrow$ (q), (C) $\rightarrow$ (p), (D) $\rightarrow$ (r)

6. C

The potential difference between the two spheres depends on the charge on the inner sphere only. Hence, the potential difference will remain unchanged because by earthing the sphere B charge on A remains constant. Let V'_A be the new potential at A. Then

$$V_A - V_B = V'_A - V'_B$$

But, $V'_B = 0$, as it is earthed,

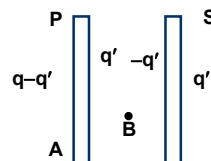
$$\text{Hence, } V'_A = V_A - V_B$$

7. B

8. $\frac{q}{2A\epsilon_0}$

Charge distribution will be as shown

$$\text{Field at point B} = \frac{q - q'}{2A\epsilon_0} + \frac{q'}{2A\epsilon_0} + \frac{q'}{2A\epsilon_0} - \frac{q'}{2A\epsilon_0} = \frac{q}{2A\epsilon_0}$$

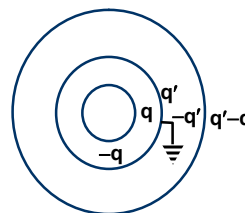


9. From Gauss' Law, inner surface of B must have charge $-q$ let charge on outer surface of B be q' . Then, remaining charge distribution must be as shown.

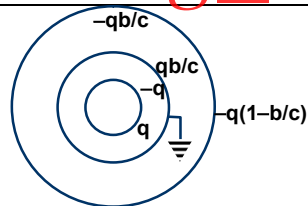
Potential on shell B = 0

$$\Rightarrow \frac{q}{4\pi \epsilon_0 b} + \frac{q' - q}{4\pi \epsilon_0 b} - \frac{q}{4\pi \epsilon_0 c} = 0$$

$$\Rightarrow q' = q \frac{b}{c}$$



Thus, charge on various surfaces should be as shown in figure below.



10. $\frac{2Q}{4\pi\epsilon_0 R} + \frac{q}{4\pi\epsilon_0 R}$

11. $1 : 4 : 9$

12. C

13. $\frac{1}{4\pi\epsilon} \left(\frac{Q + q/3}{R} \right)$

Total potential at C = potential at A